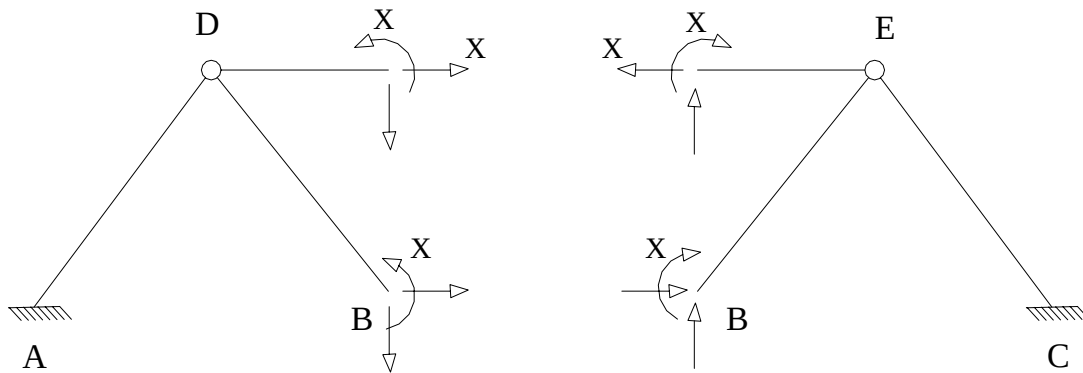
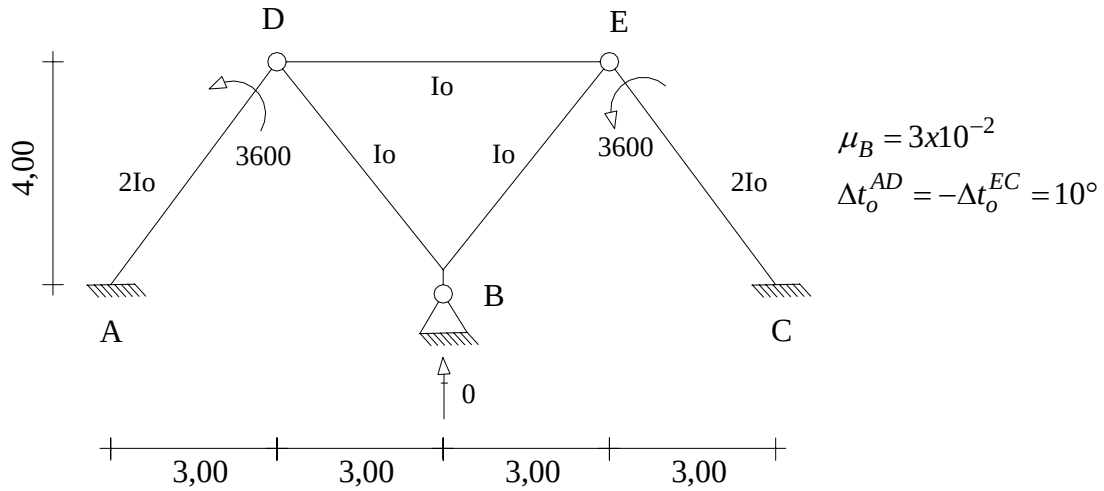


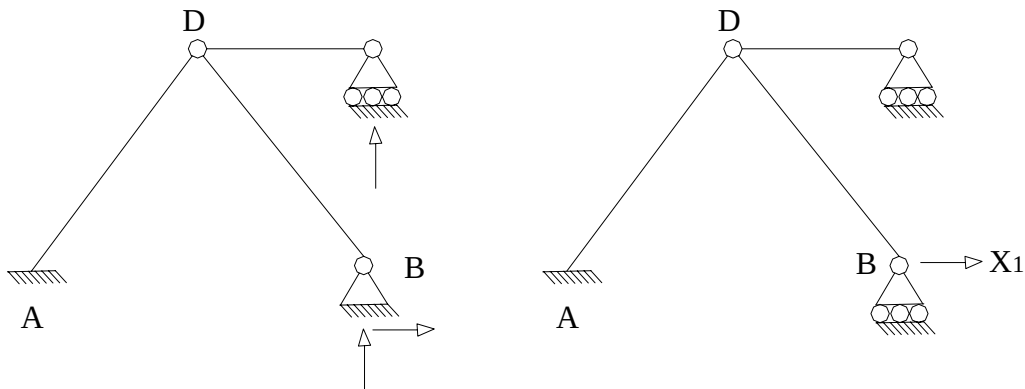
CAPITULO V

MÉTODO DE LAS FUERZAS

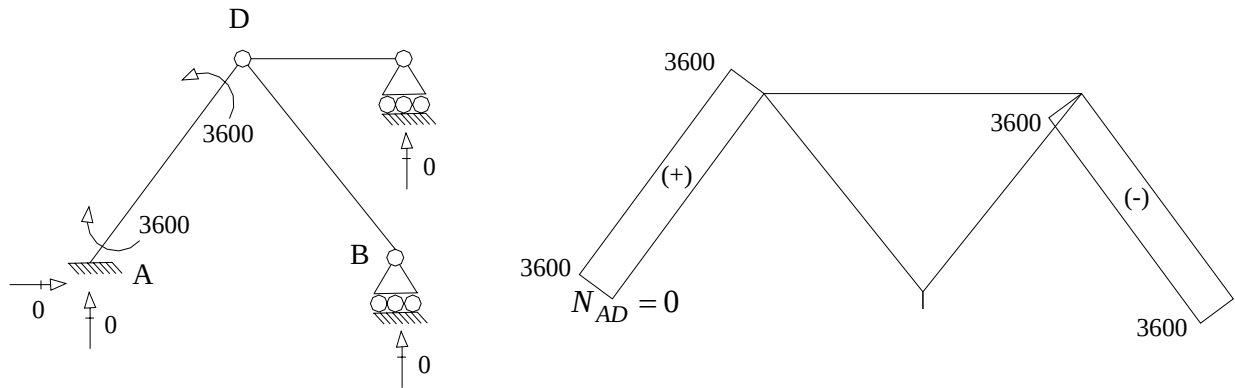
1.- Estructuras Simétricas con Solicitaciones Antisimétricas.



Sistema Primario:



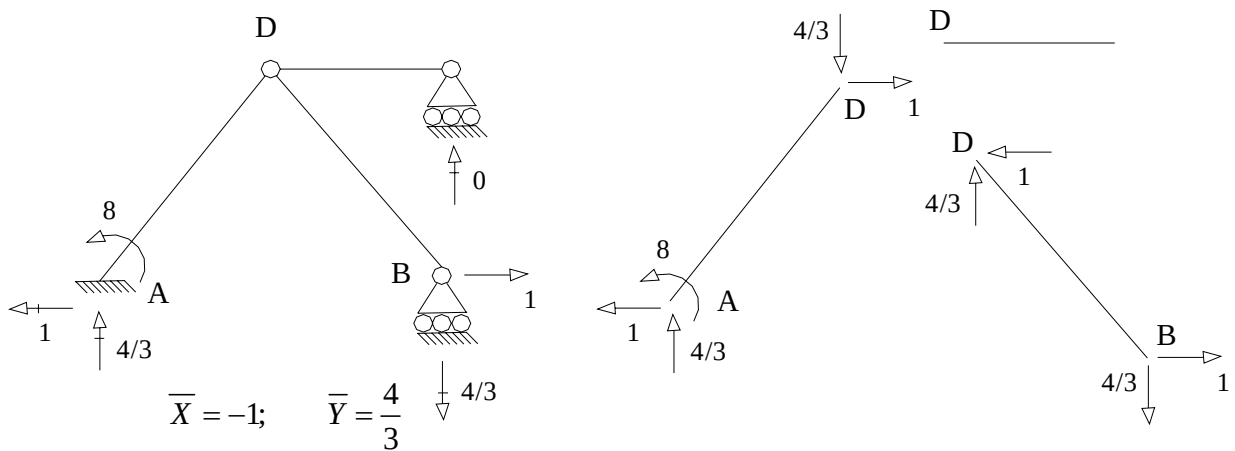
Sistema Cero:



Sistema Uno:

$$+\circlearrowleft \sum M_D^{DB} = 0 \Rightarrow -4x1 + 3B_v = 0 \Rightarrow B_v = \frac{4}{3}$$

$$+\circlearrowleft \sum M_D^{DA} = 0 \Rightarrow M_A 3x \frac{4}{3} + 4 = 0 \Rightarrow M_A = -8$$



$$N_{AD} = 1x \frac{3}{5} - \frac{4}{3} \left(\frac{4}{5} \right) = \frac{3}{5} - \frac{16}{15} = \frac{-7}{15} // \Omega = \frac{-7}{3}$$

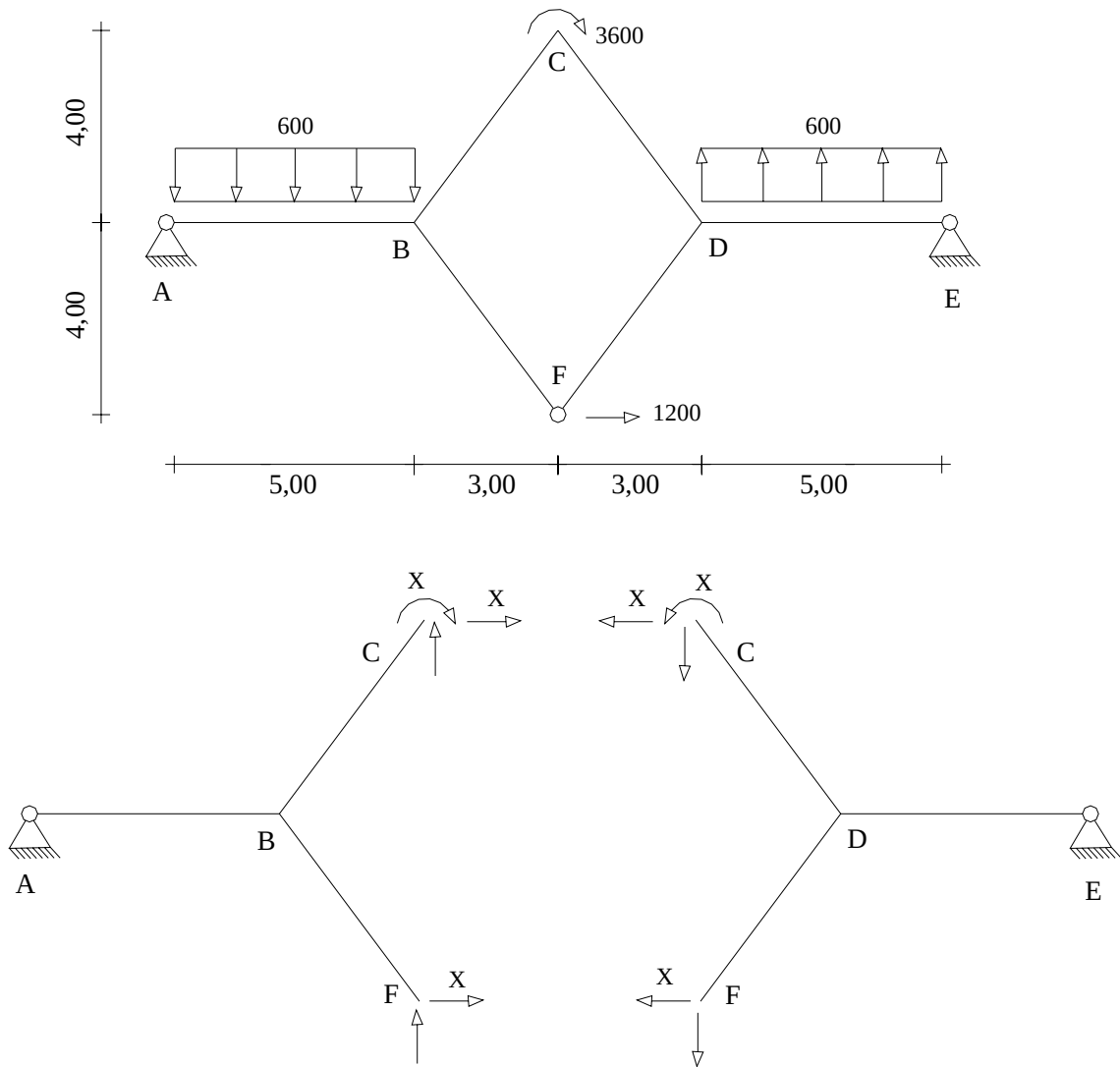
$$D_{10} = \left(\frac{-1}{2} x \frac{8x3600x5}{2EI_o} \right) x 2 - \left(\frac{7}{3} x 10x10^{-5} x \frac{10^6}{EI_o} \right) x 2 = \frac{(-72466,67)}{EI_o}$$

$$d_{11} = \left(\frac{1}{3} x \frac{8x8x5}{2EI_o} \right) x 2 = \frac{106,66}{EI_o}$$

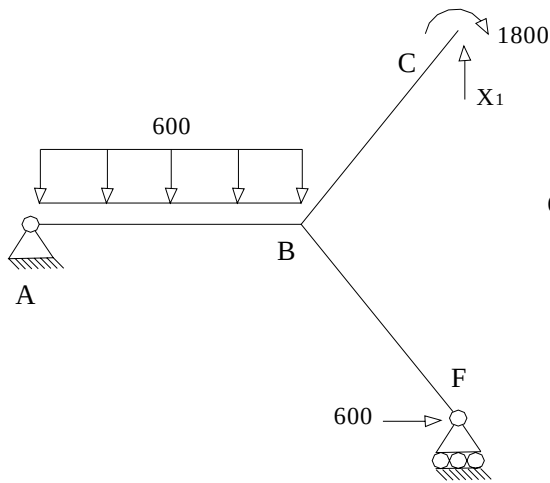
$$D_1 = D_{10} + X_1 d_{11} \Rightarrow 3 \times 10^{-2} x \frac{10^6}{EI_o} = \frac{-72466,67}{EI_o} + X_1 \left(\frac{106,66}{EI_o} \right)$$

$$X_1 = 960,60$$

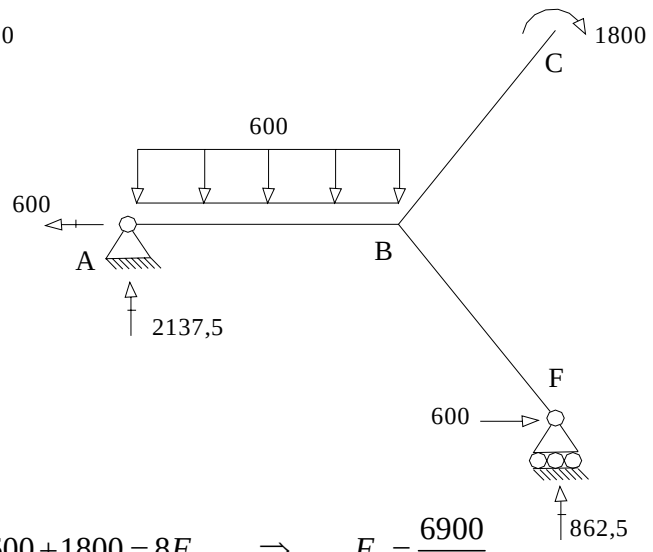
2.- Estructuras Simétricas con Solicitaciones Antisimétricas.



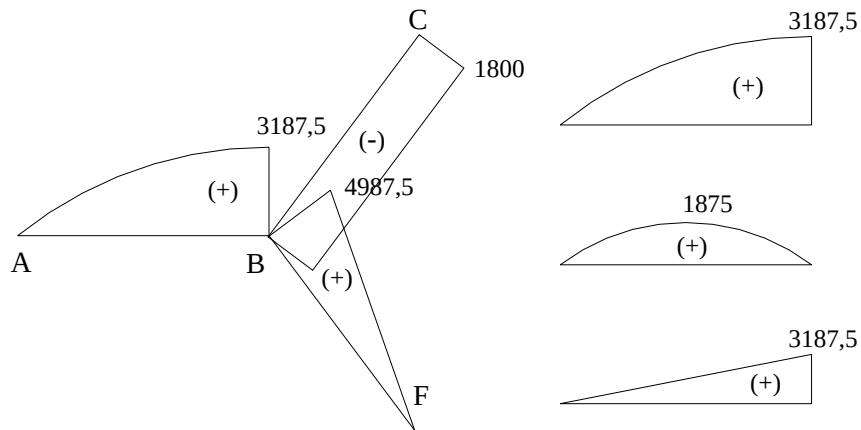
Sistema Primario:



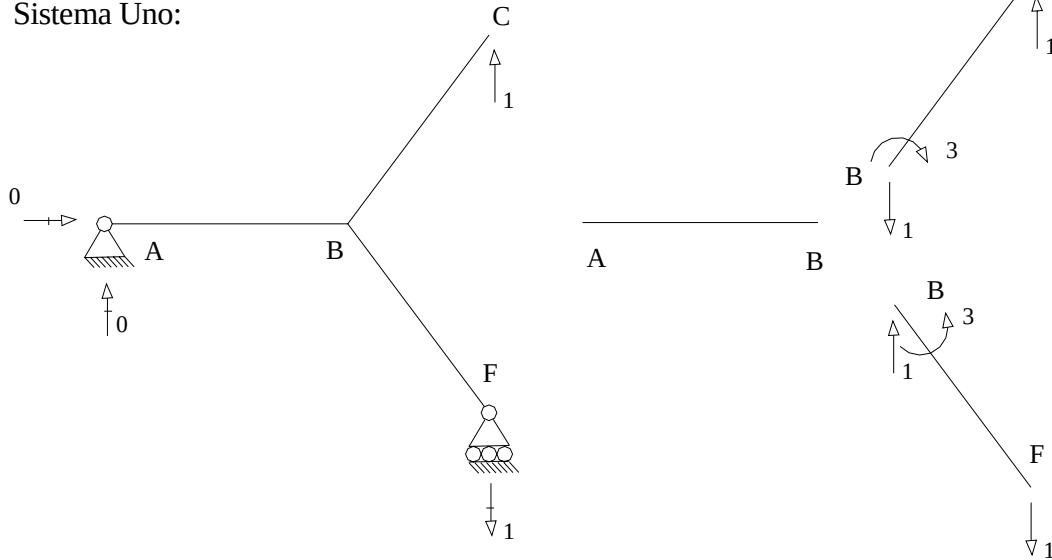
Sistema Cero:



$$+\circlearrowleft \sum M_A = 0 \Rightarrow 5 \times 600 \times \frac{5}{2} - 4 \times 600 + 1800 = 8F_v \Rightarrow F_v = \frac{6900}{8}$$



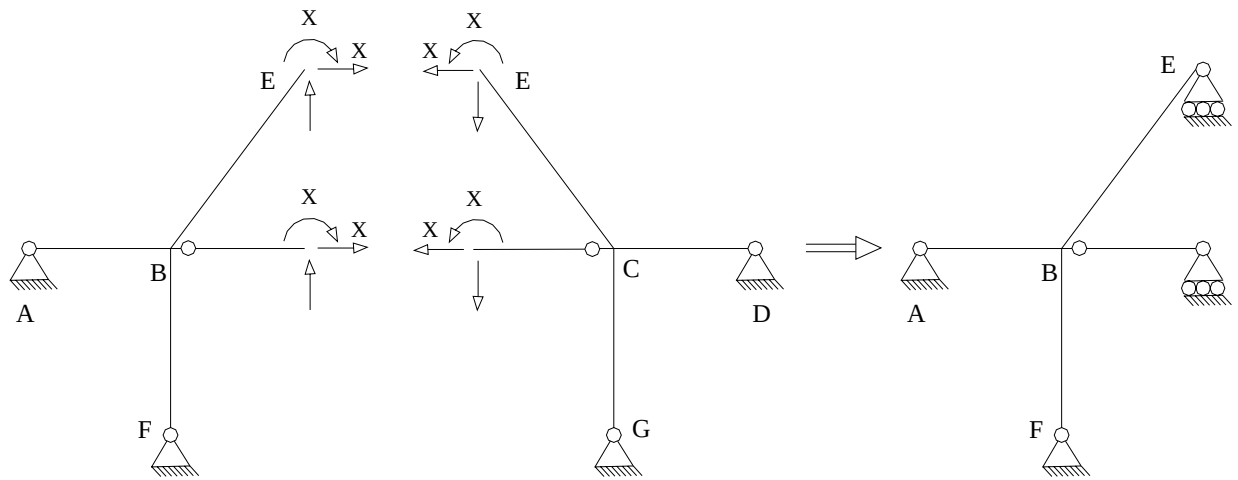
Sistema Uno:



$$d_{11} = \frac{1}{3EI_o} (3 \times 3 \times 5 + 3 \times 3 \times 5) = \frac{30}{EI_o}$$

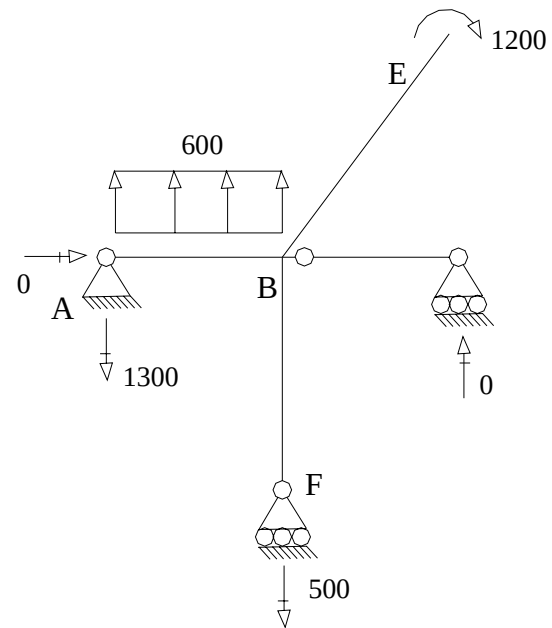
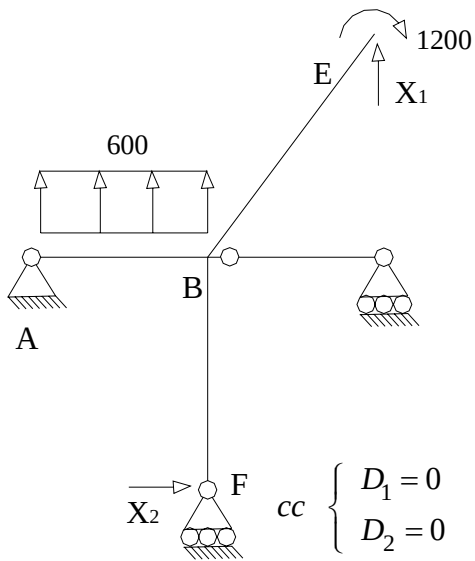
3.- Estructuras Simétricas con Solicitaciones Antisimétricas.



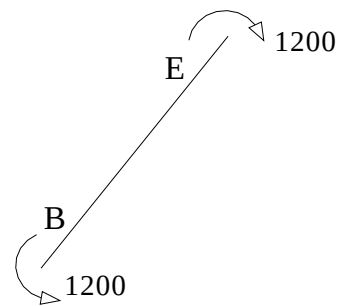
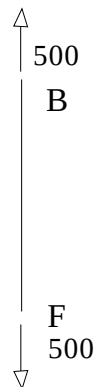
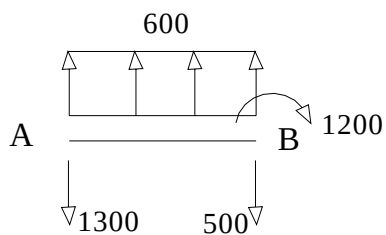


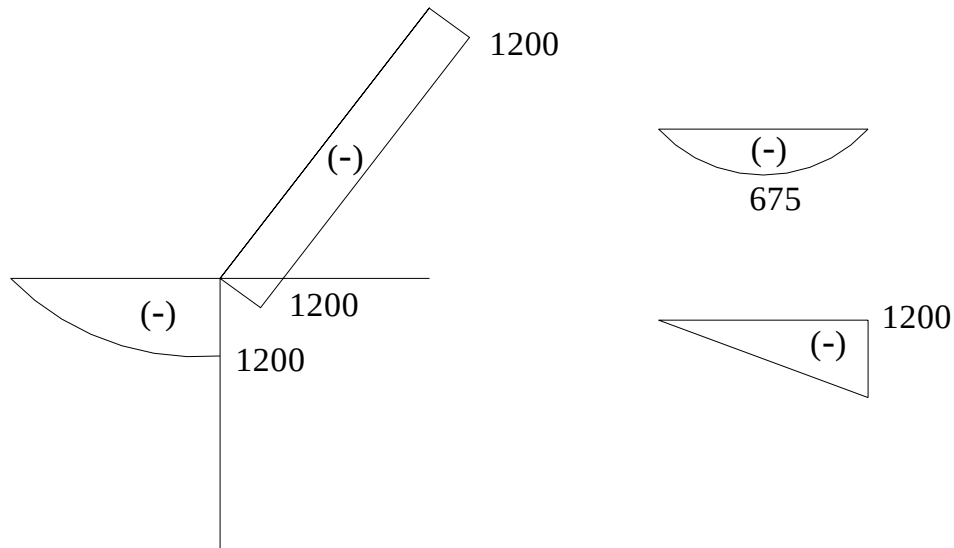
Sistema Primario:

Sistema Cero:



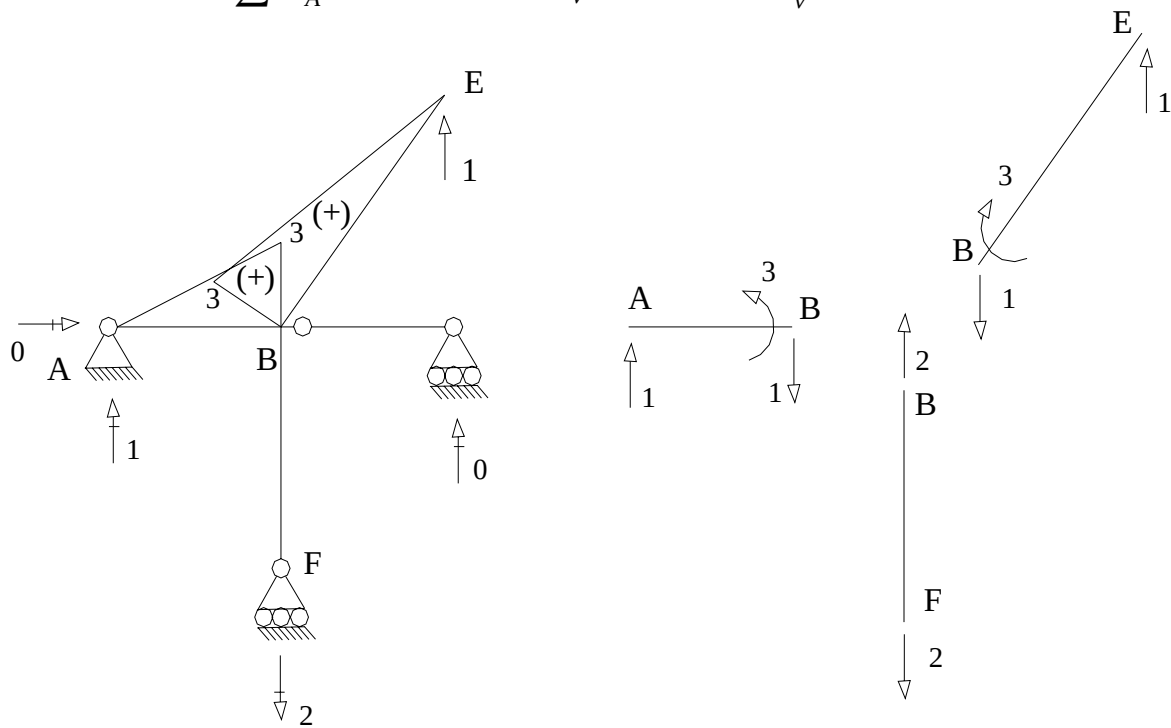
$$+\circlearrowleft \sum M_A = 0 \Rightarrow -2700 + 1200 + 3F_v = 0 \Rightarrow F_v = 500$$





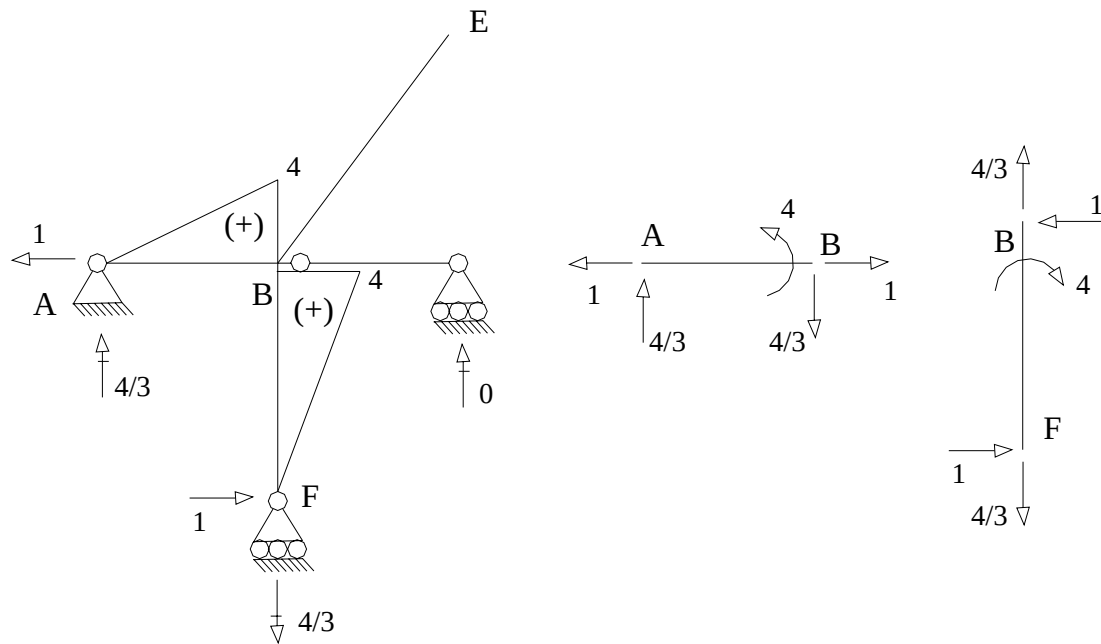
Sistema Uno:

$$+\circlearrowleft \sum M_A = 0 \Rightarrow -6 + 3F_v = 0 \Rightarrow F_v = 2$$



Sistema Dos:

$$+\circlearrowleft \sum M_A = 0 \Rightarrow -4 + 3F_v = 0 \Rightarrow F_v = \frac{4}{3}$$



$$D_{10} = \frac{-1}{2} \times \frac{1200 \times 3 \times 5}{EI_o} - \frac{1}{3} \times \frac{675 \times 3 \times 3}{2EI_o} - \frac{1}{3} \times \frac{1200 \times 3 \times 3}{2EI_o} = \frac{-11812,5}{EI_o}$$

$$D_{20} = \frac{-1}{3} \times \frac{675 \times 4 \times 3}{2EI_o} - \frac{1}{3} \times \frac{4 \times 1200 \times 3}{2EI_o} = \frac{-3750}{EI_o}$$

$$d_{11} = \frac{1}{3} \left(\frac{3^3}{2EI_o} + \frac{3^2 \times 5}{EI_o} \right) = \frac{19,5}{EI_o}$$

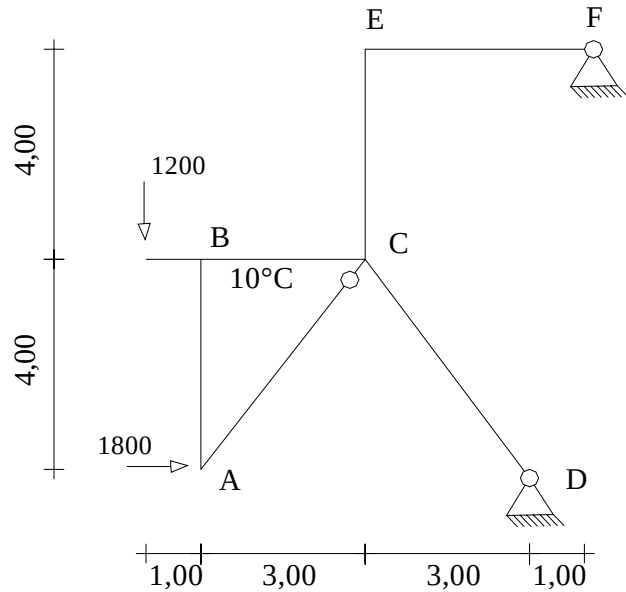
$$d_{12} = d_{21} = \frac{1}{3(2EI_o)} \times 3 \times 4 \times 3 = \frac{6}{EI_o}$$

$$11812,5 = X_1(19,5) + X_2(6)$$

$$3750 = X_1(6) + X_2(18,66)$$

$$X_1 = 603,66; \quad X_2 = 6,86$$

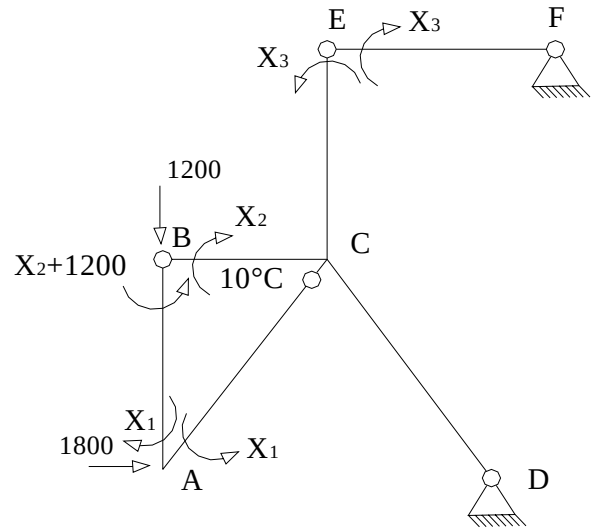
4.-



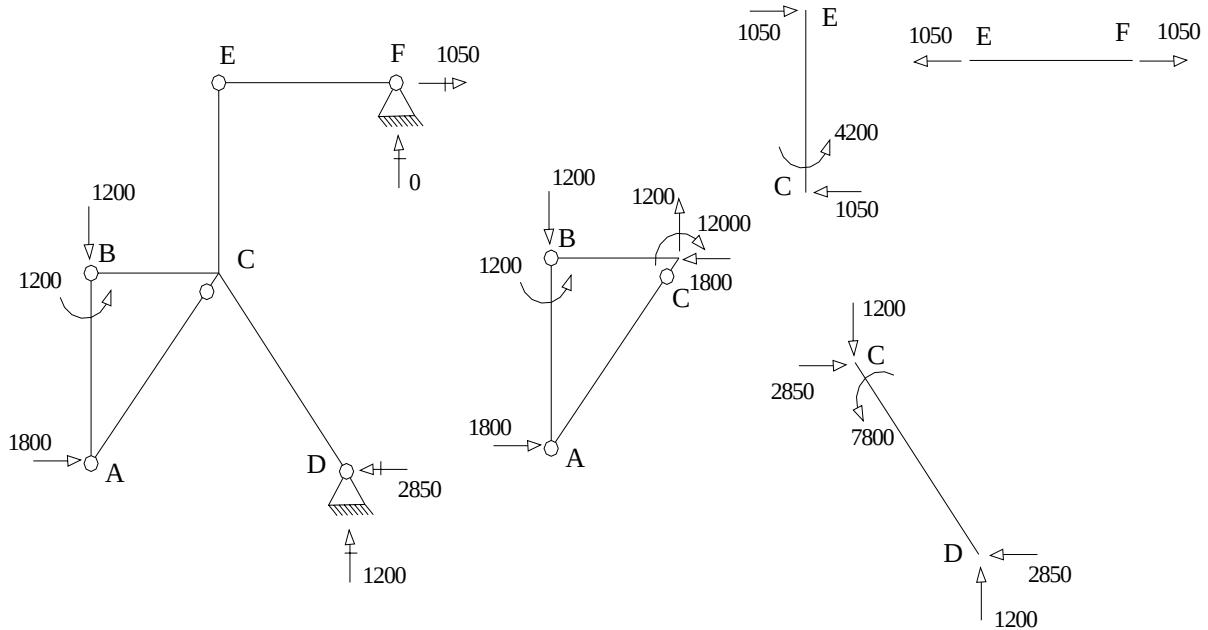
$$h_{BC} = 0,50\text{cm}$$

$$\mu_F = 2 \times 10^{-2} m$$

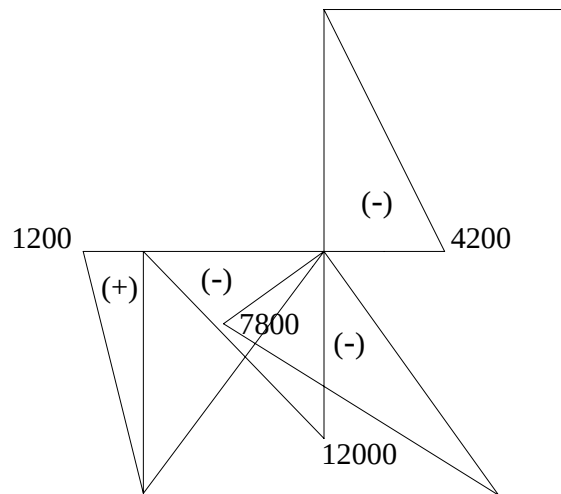
Sistema Primario:



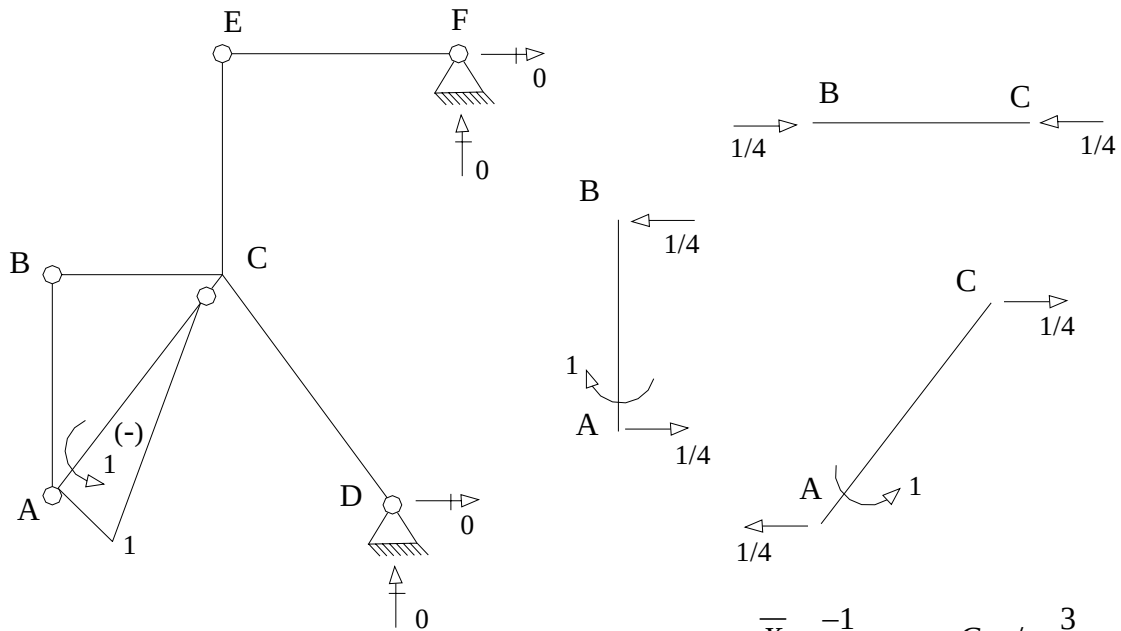
Sistema Cero:



$$+\circlearrowleft \sum M_D = 0 \Rightarrow -6 \times 1200 - 1200 + 8F_h = 0 \Rightarrow F_h = 1050$$



Sistema Uno:



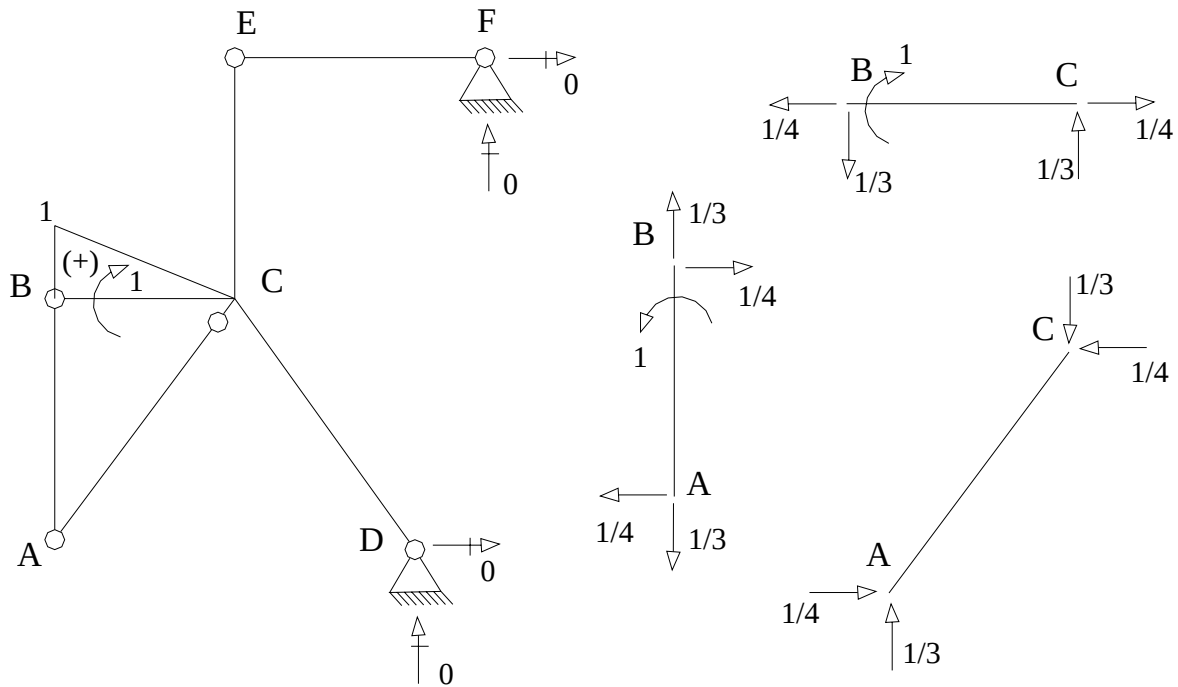
$$N_{EC} = 0; \quad N_{EF} = -\frac{1}{4} // \Omega = -\frac{3}{4}$$

$$N_{CB} = \frac{3}{20} // \Omega = \frac{3}{20} (5) = \frac{3}{4}$$

$$\bar{X} = \frac{-1}{4}; \quad \cos \phi = \frac{3}{5}$$

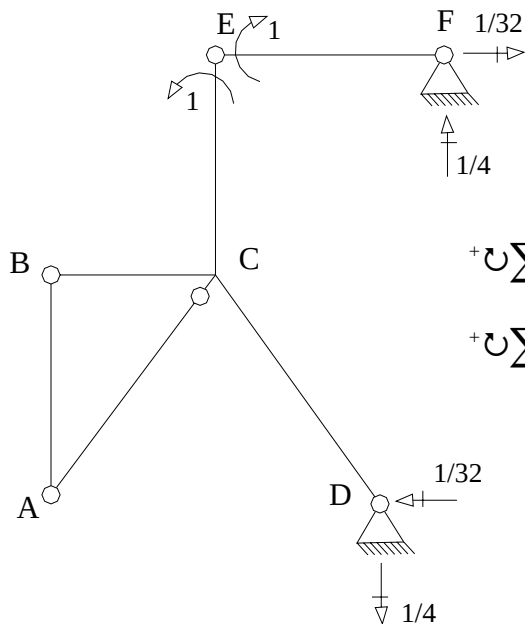
$$N_{AC} = \frac{1}{4} \left(\frac{3}{5} \right) = \frac{3}{20}$$

Sistema Dos:



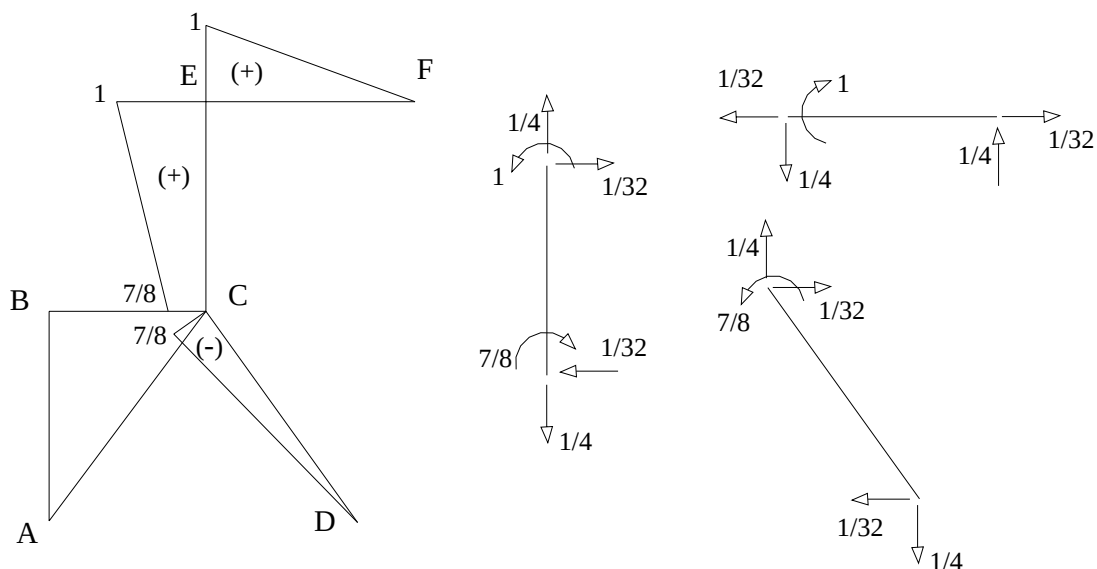
$$N_{BC} = \frac{1}{4} // \Omega = \frac{3}{4}; \quad N_{AB} = \frac{1}{3} // \Omega = \frac{4}{3}; \quad N_{AC} = \frac{-5}{12} // \Omega = -\frac{25}{12}$$

Sistema Tres:



$$+\circlearrowleft \sum M_E = 0 \Rightarrow 1 - 4F_v = 0 \Rightarrow F_v = \frac{1}{4}$$

$$+\circlearrowleft \sum M_D = 0 \Rightarrow 8F_h - \frac{1}{4} = 0 \Rightarrow F_h = \frac{1}{32}$$



$$D_{10} = \frac{1}{6} \times \frac{1200 \times 4}{EI_o} + \left(-\frac{1}{4} - \frac{3}{4}\right) \times 10^{-5} \times 5 \times \frac{10^6}{EI_o} + \left[\frac{4}{2} \times (10) - \frac{5}{2} \times (-10)\right] \times 10^{-5} \times 0,5 \times \frac{10^6}{EI_o} = \frac{183}{EI_o}$$

$$D_{20} = \frac{1}{3} \times \frac{1200 \times 4}{EI_o} + \left(\frac{3}{4} - \frac{4}{3} - \frac{25}{12}\right) \times 10^{-5} \times 5 \times \frac{10^6}{EI_o} + \left(\frac{3}{2} \times 10 + \frac{4}{2} \times 10\right) \times 10^{-5} \times 0,5 \times \frac{10^6}{EI_o} = \frac{1641,66}{EI_o}$$

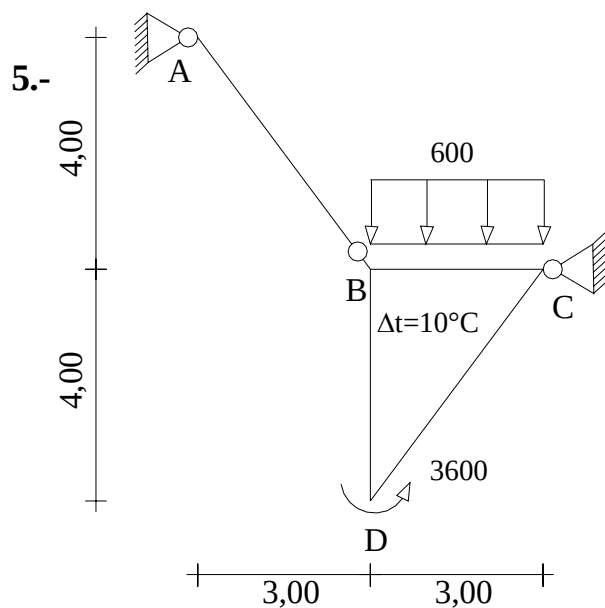
$$d_{11} = \frac{1}{3EI_o} (4 + 5) = \frac{3}{EI_o};$$

$$d_{11} = \frac{1}{3EI_o} (4 + 3) = \frac{7}{3EI_o}$$

$$d_{12} = d_{21} = \frac{1}{6} \times \frac{4}{EI_o} = \frac{2}{3EI_o};$$

$$d_{13} = d_{31} = 0;$$

$$d_{23} = d_{32} = 0$$



$$h = \frac{1}{2};$$

$$\Delta(\Delta t) = \Delta t_i - \Delta t_s$$

$$\alpha t = 1 \times 10^{-6};$$

$$\Delta t_o = \frac{\Delta t_2 + \Delta t_3}{2}$$

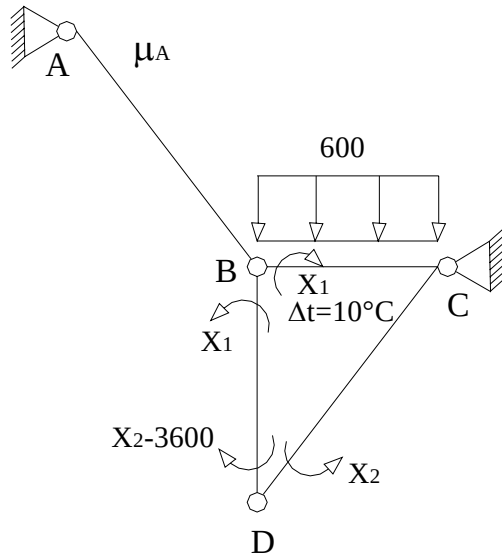
$$\mu_A = 2 \times 10^{-2};$$

$$\Delta(\Delta t)_{BC} = 10^\circ\text{C}$$

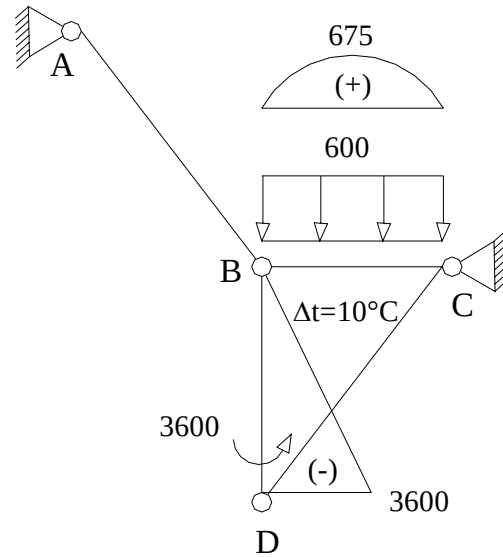
$$\Delta(\Delta t)_{CD} = -10^\circ\text{C};$$

$$\Delta(\Delta t)_{BD} = 10^\circ\text{C}$$

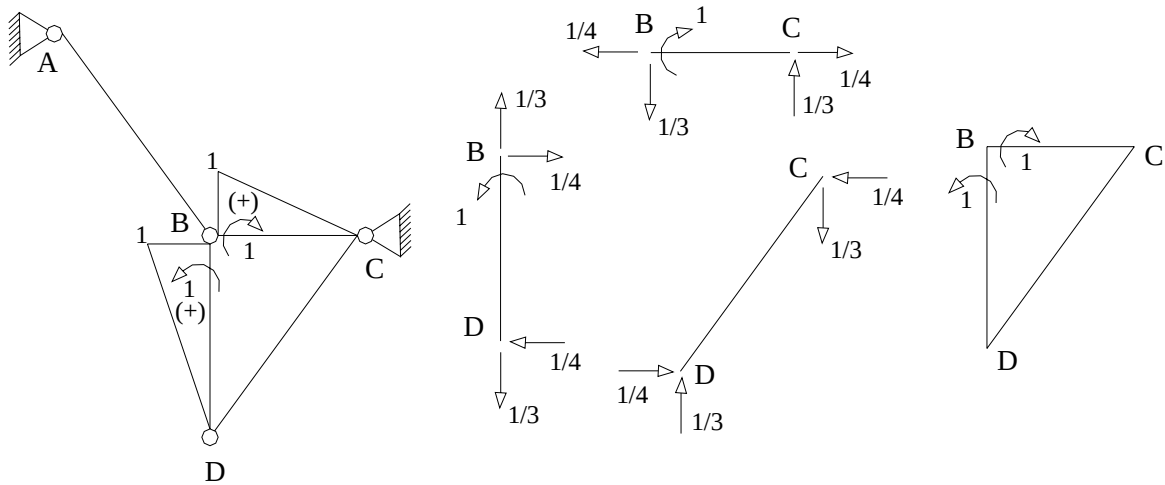
Sistema Primario:



Sistema Cero:



Sistema Uno:



$$\bar{X} = \frac{1}{4}; \quad \cos \phi = \frac{3}{5}$$

$$\bar{Y} = \frac{1}{3}; \quad \cos \phi = \frac{4}{5}$$

$$N_{DC} = -\frac{1}{4} \left(\frac{3}{5} \right) - \frac{1}{3} \left(\frac{4}{5} \right)$$

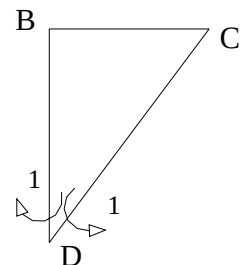
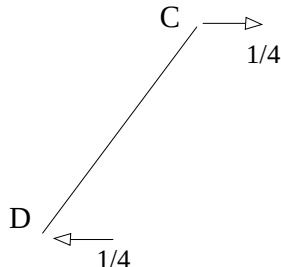
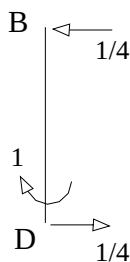
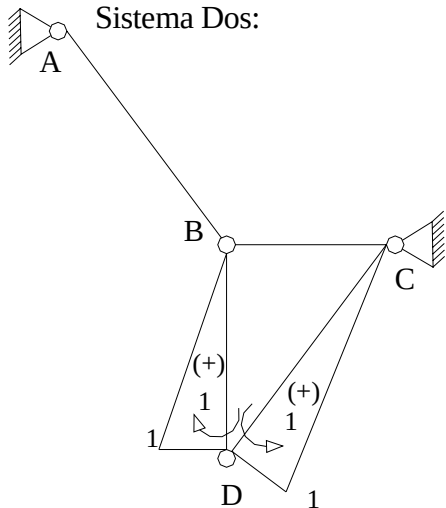
$$N_{DC} = -\frac{5}{12}$$

$$N_{BC} = \frac{1}{4} // \Omega = \frac{3}{4}$$

$$N_{BD} = \frac{1}{3} // \Omega = \frac{4}{3}$$

$$N_{CD} = \frac{-5}{12} // \Omega = \frac{-25}{12}$$

Sistema Dos:



$$3x \frac{1}{4} // \Omega_{BC} = -\frac{3}{4}$$

$$5x \frac{3}{20} // \Omega_{DC} = \frac{3}{4}$$

$$\bar{X} = -\frac{1}{4}; \bar{Y} = 0$$

$$N_{DC} = \frac{1}{4} \left(\frac{3}{5} \right) = \frac{3}{20}$$

$$D_{10} = -Tr_1 - \frac{1}{6} x \frac{3600x4}{EI_o} + \frac{1}{3} x \frac{675x3}{EI_o} + \left(\frac{3}{4} + \frac{4}{3} - \frac{25}{12} \right) x 10^{-5} x 5x \frac{10^6}{EI_o} +$$

$$+ \left(\frac{3}{2} x 10 + \frac{4}{2} x 10 \right) x 10^{-5} x 2x \frac{10^6}{EI_o} = \frac{-1025}{EI_o}$$

$$D_{20} = -Tr_2 - \frac{1}{3} x \frac{3600x4}{EI_o} + \left(-\frac{3}{4} + \frac{3}{4} + 0 \right) x 10^{-5} x 5x \frac{10^6}{EI_o} +$$

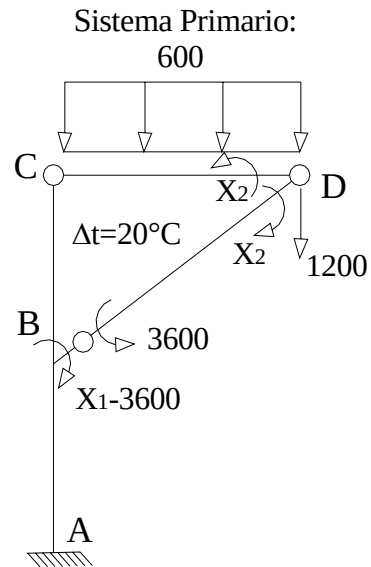
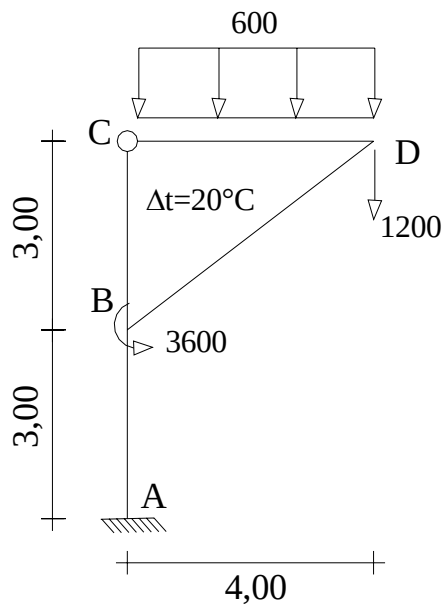
$$+ \left[\frac{4}{2} x 10 - \frac{5}{2} x (-10) \right] x 10^{-5} x 2x \frac{10^6}{EI_o} = \frac{-3900}{EI_o}$$

$$d_{11} = \frac{1}{3EI_o} (4+3) = \frac{7}{3EI_o}; \quad d_{22} = \frac{1}{3EI_o} (4+5) = \frac{3}{EI_o}$$

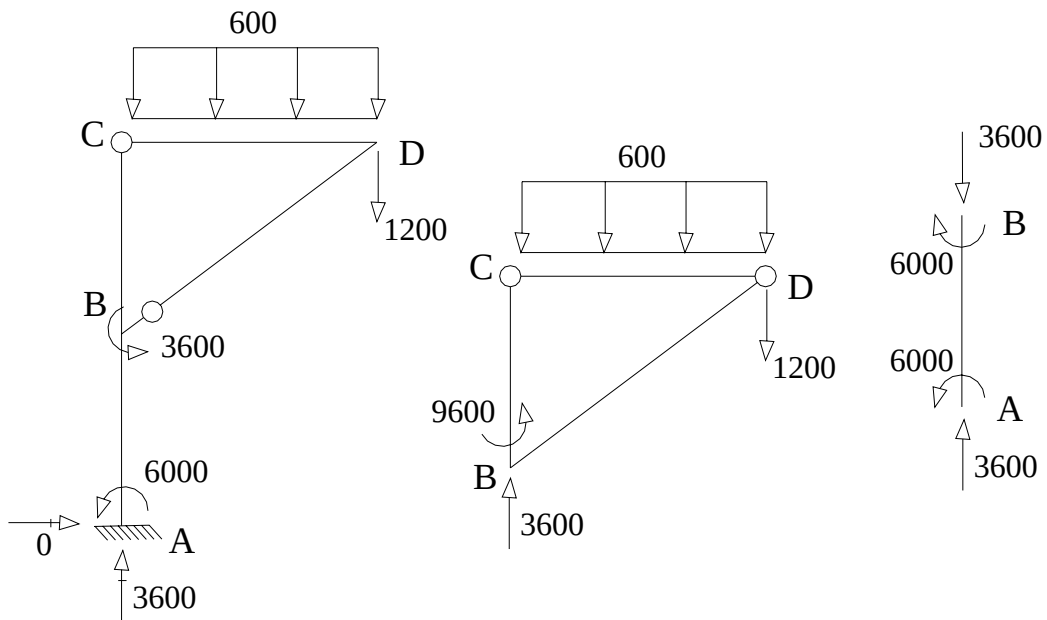
$$d_{12} = d_{21} = \frac{1}{6} x \frac{4}{EI_o} = \frac{2}{3EI_o}$$

$$\begin{aligned} 0 &= -1025 + X_1 x \frac{7}{3} + X_2 x \frac{2}{3} \\ 0 &= -3900 + X_1 x \frac{2}{3} + X_2 x 3 \end{aligned} \Rightarrow \begin{cases} X_1 = 72,46 \\ X_2 = 1283,90 \end{cases}$$

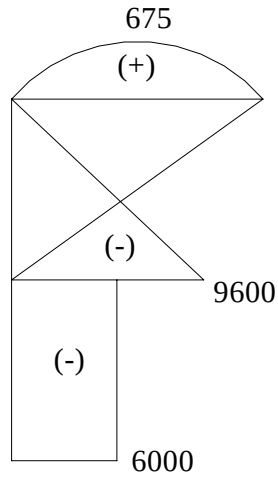
6.-



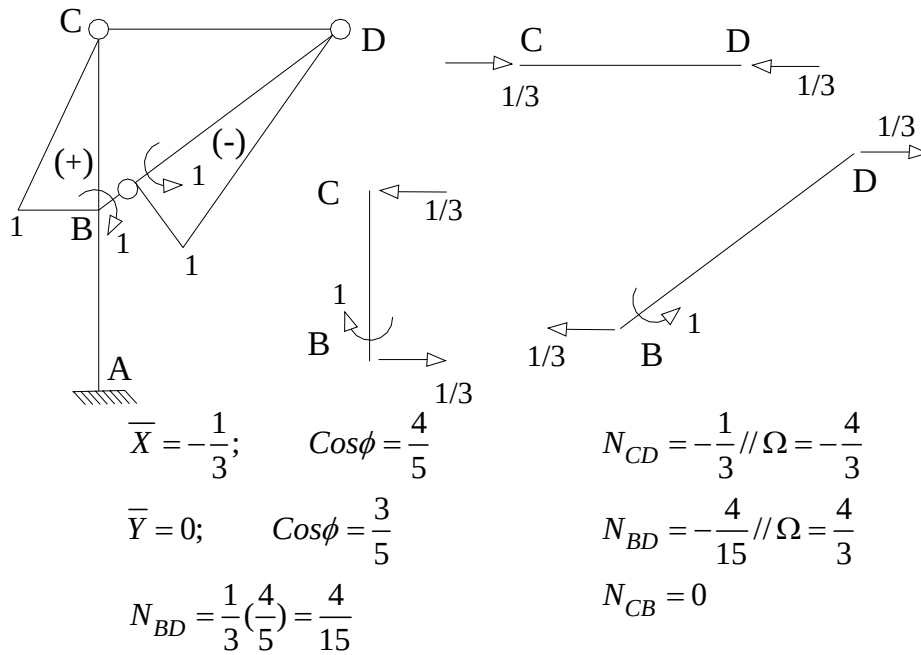
Sistema Cero:



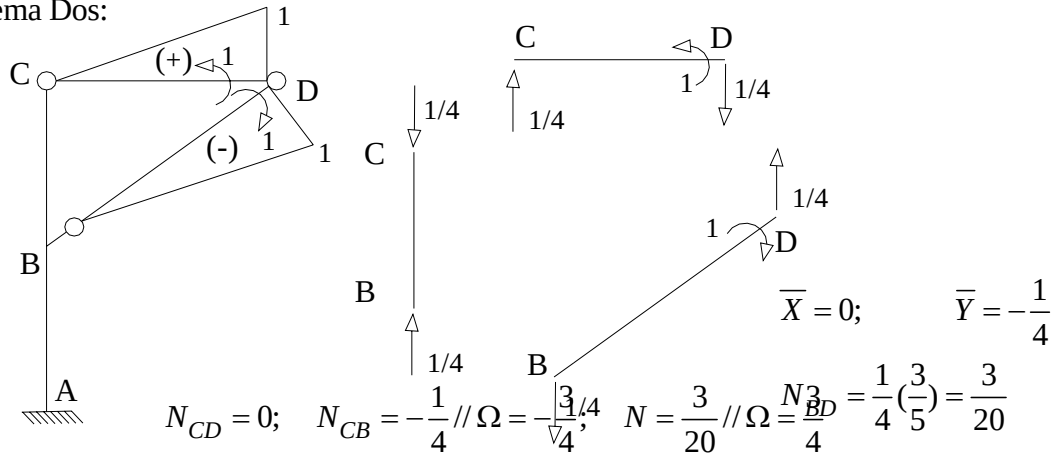
$$+\circlearrowleft \sum M_A = 0 \Rightarrow -3600 + 4 \times 600 \times 2 + 4 \times 1200 + M_A = 0 \Rightarrow M_A = -6000$$



Sistema Uno:



Sistema Dos:



$$D_{10} = -\frac{1}{3} \times \frac{9600 \times 3}{EI_o} + \left(-\frac{4}{3} + \frac{4}{3} + 0\right) \times 10^{-5} \times \frac{10^6}{EI_o} \times 10 +$$

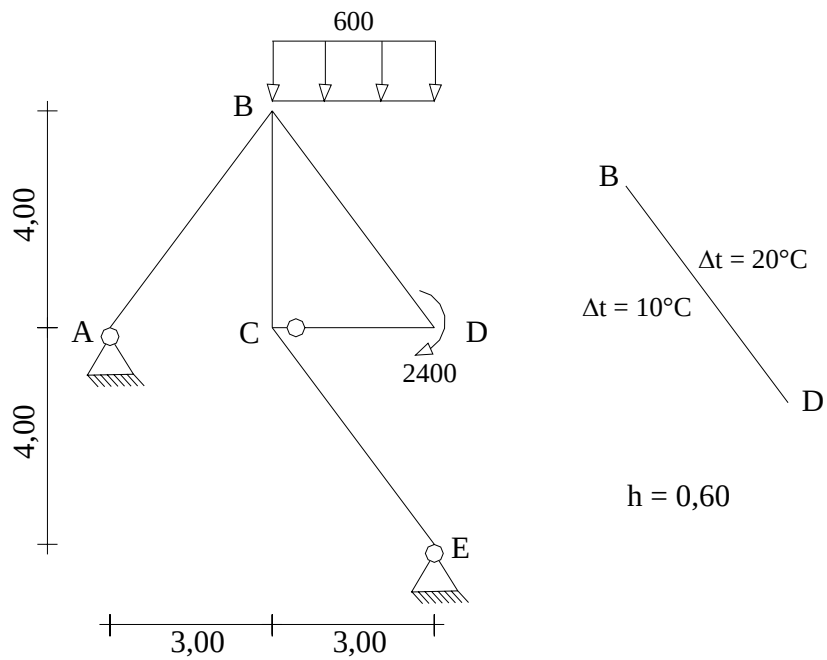
$$+ \left[\frac{3}{2} \times 20 - \frac{5}{2} \times (-20)\right] \times 10^{-5} \times 2 \times \frac{10^6}{EI_o} = -\frac{8000}{EI_o}$$

$$D_{20} = \frac{1}{3} \times \frac{675 \times 4}{EI_o} + \left(-\frac{3}{4} + \frac{3}{4}\right) \times 10^{-5} \times \frac{10^6}{EI_o} \times 10 +$$

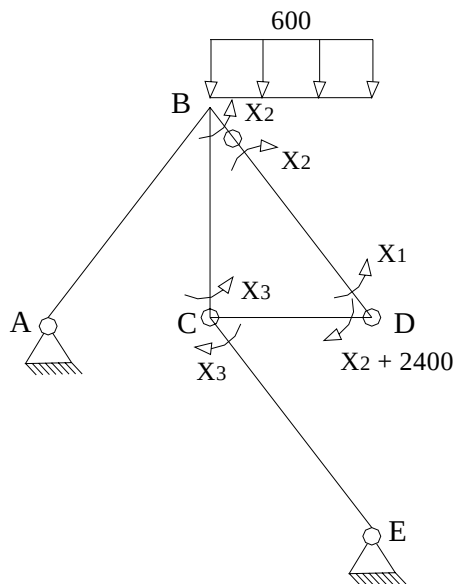
$$+ \left[\frac{4}{2} \times 20 - \frac{5}{2} \times (-20)\right] \times 10^{-5} \times 2 \times \frac{10^6}{EI_o} = \frac{2700}{EI_o}$$

$$d_{11} = (3+5) \frac{1}{3EI_o} = \frac{8}{3EI_o}; \quad d_{22} = \frac{1}{3EI_o} (4+5) = \frac{3}{EI_o}$$

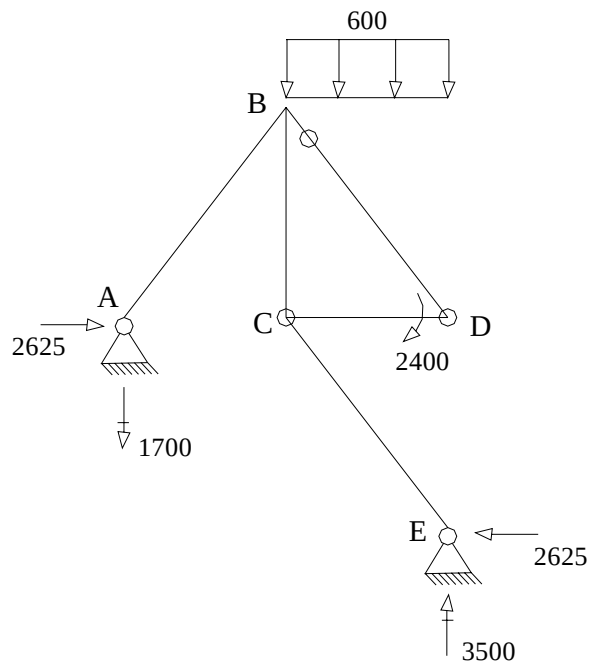
7.-



Sistema Primario:

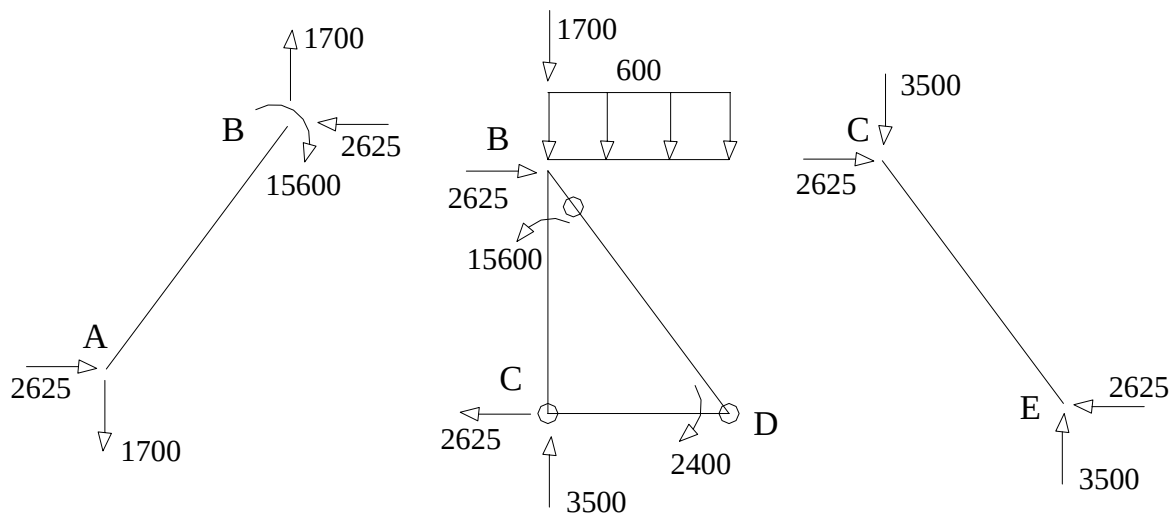


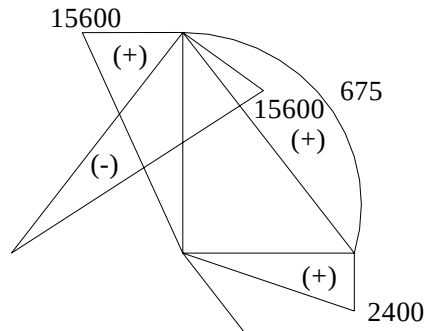
Sistema Cero:



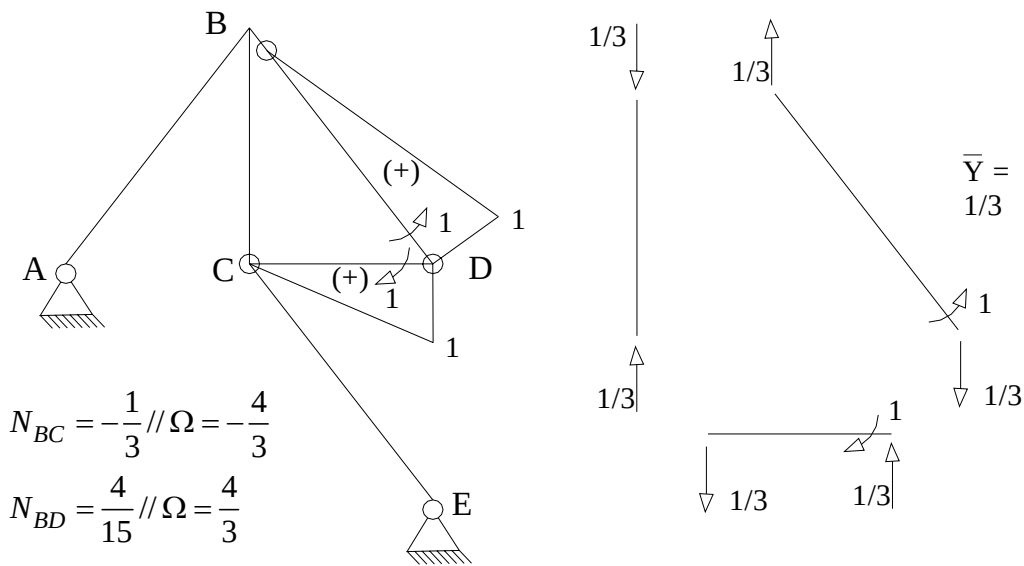
$$+\circlearrowleft \sum M_C^{ABCD} = 0 \Rightarrow 2400 + 600 \times 3 \times \frac{3}{2} + 3A_v = 0 \Rightarrow A_v = -1700$$

$$+\circlearrowleft \sum M_C^{CE} = 0 \Rightarrow -3 \times 3500 - 4A_h = 0 \Rightarrow A_h = -2625$$

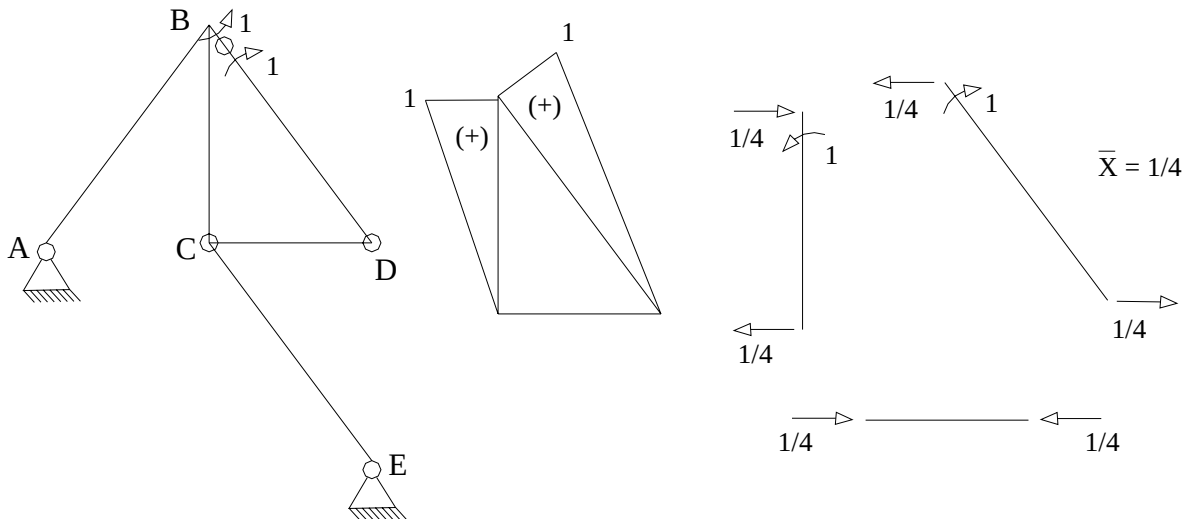




Sistema Uno:



Sistema Dos:

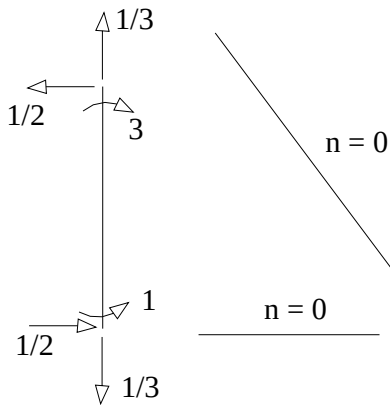
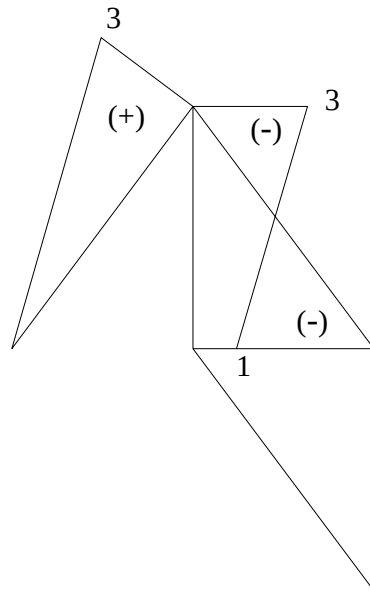
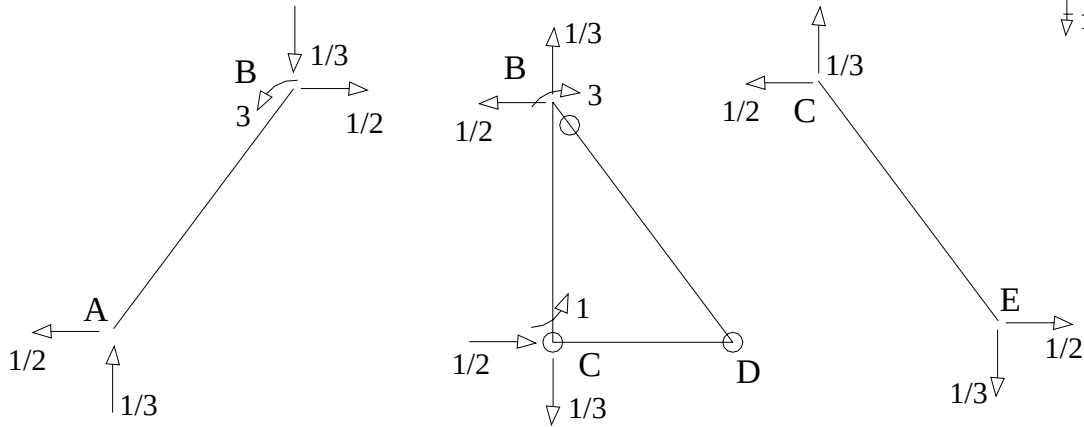
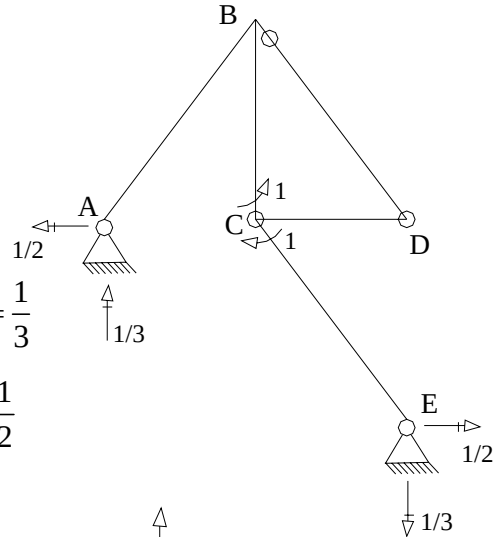


$$N_{CD} = -\frac{1}{4} // \Omega = -\frac{3}{4}; \quad N_{BD} = \frac{3}{20} // \Omega = \frac{3}{4}$$

Sistema Tres:

$$+\circlearrowleft \sum M_C^{ABCD} = 0 \Rightarrow -1 + 3A_v = 0 \Rightarrow A_v = \frac{1}{3}$$

$$+\circlearrowleft \sum M_C^{CE} = 0 \Rightarrow 1 + 1 - 4E_h = 0 \Rightarrow E_h = \frac{1}{2}$$



$$D_{10} = \frac{1}{3} \times \frac{675 \times 5}{EI_o} + \frac{1}{3} \times \frac{2400 \times 3}{EI_o} + \frac{4}{3} \times 15 \times 10^{-5} \times \frac{10^6}{EI_o} + \frac{5}{2} \times (-10) \times 10^{-5} \times \frac{1}{0,60} \times \frac{10^6}{EI_o}$$

$$D_{10} = \frac{3308,33}{EI_o}$$

$$D_{20} = \frac{1}{3} \times \frac{675 \times 5}{EI_o} + \frac{1}{3} \times \frac{15600 \times 4}{EI_o} + \frac{3}{4} \times 15 \times 10^{-5} \times \frac{10^6}{EI_o} + \frac{5}{2} \times (-10) \times 10^{-5} \times \frac{1}{0,60} \times \frac{10^6}{EI_o}$$

$$D_{20} = \frac{21620,83}{EI_o}$$

$$D_{30} = -\frac{1}{3} \times \frac{3 \times 15600 \times 5}{EI_o} = -\frac{78000}{EI_o}$$

$$d_{11} = (3+5) \frac{1}{3EI_o} = \frac{8}{3EI_o};$$

$$d_{22} = \frac{1}{3EI_o} (4+5) = \frac{3}{EI_o}$$

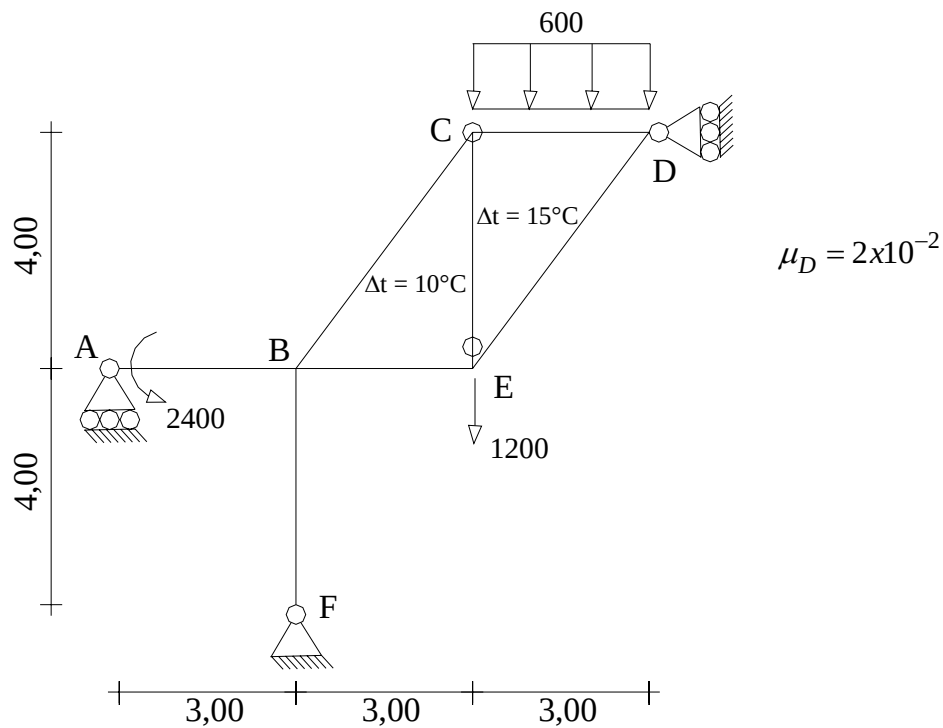
$$d_{33} = \frac{3 \times 3 \times 5}{3EI_o} + \frac{4}{6} (2 \times 3 \times 3 + 2 \times 1 \times 1 + 3 \times 1 + 1 \times 3) \times \frac{1}{EI_o} = \frac{15}{EI_o} + \frac{104}{6EI_o} = \frac{97}{3EI_o}$$

$$d_{12} = d_{21} = \frac{5}{6EI_o};$$

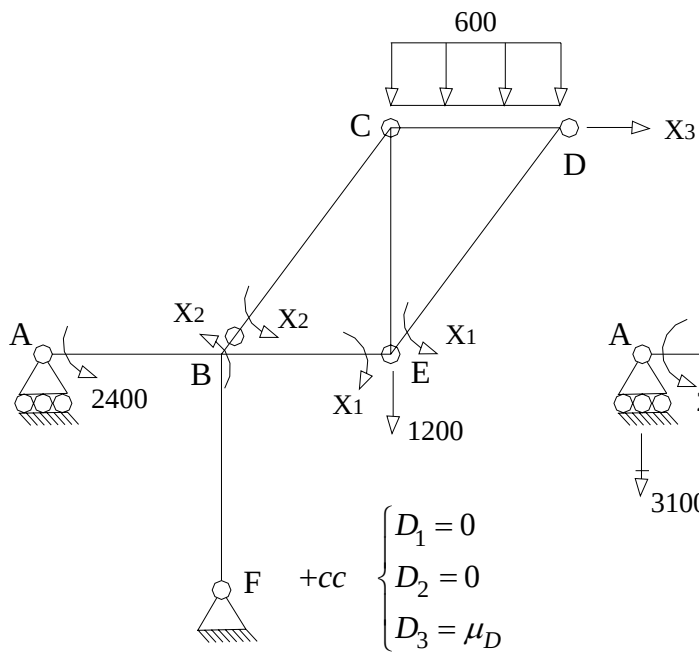
$$d_{13} = d_{31} = 0$$

$$d_{23} = d_{32} = -\frac{4}{6EI_o} (2 \times 1 \times 3 + 1 \times 1) = -\frac{28}{6EI_o} = -\frac{14}{3EI_o}$$

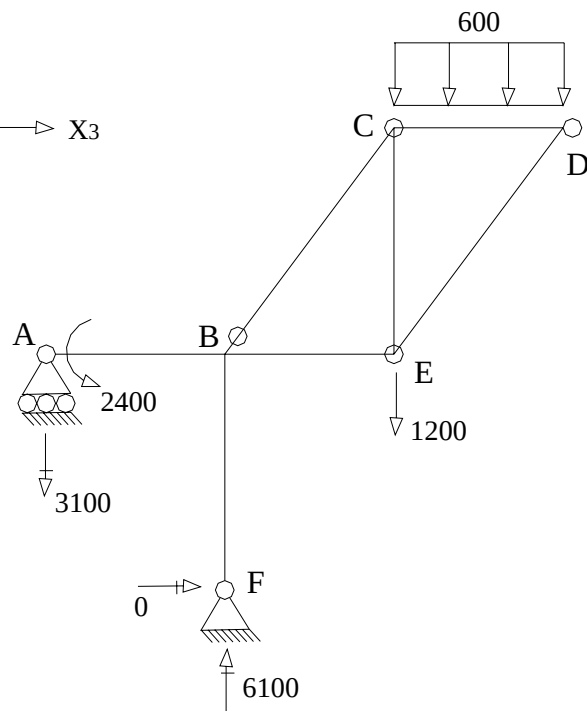
8.-



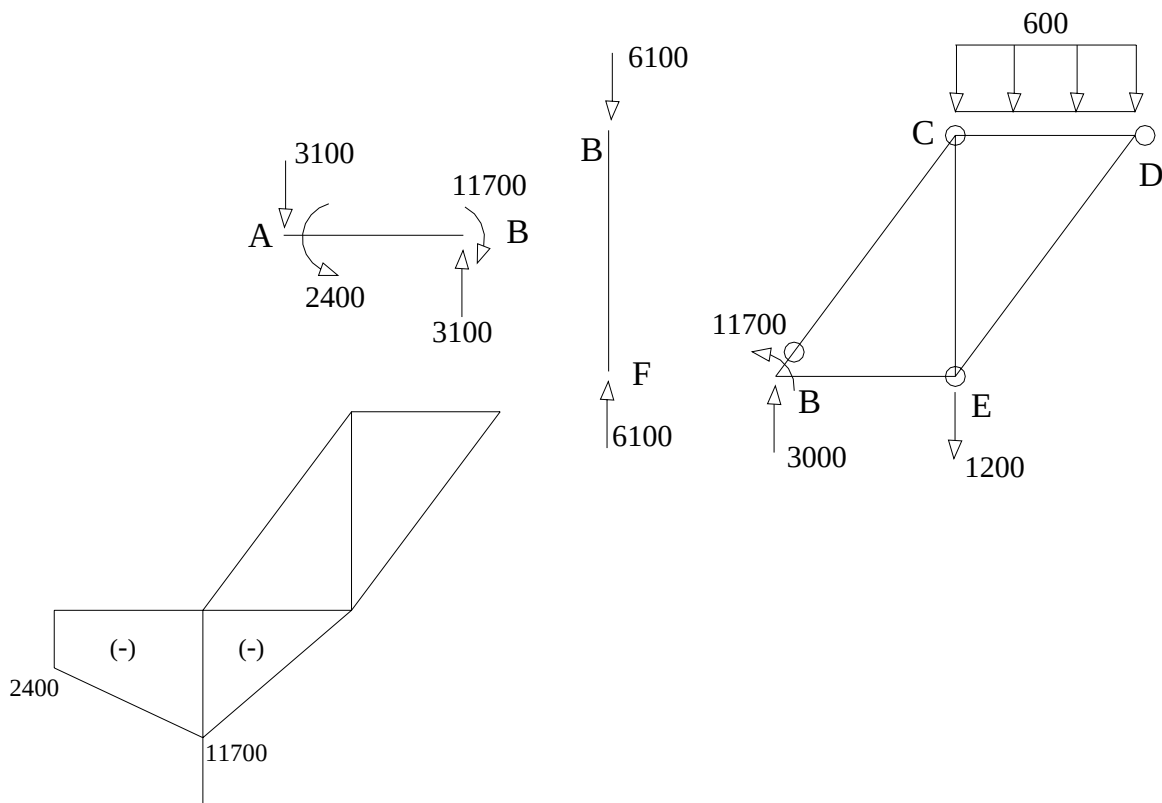
Sistema Primario:

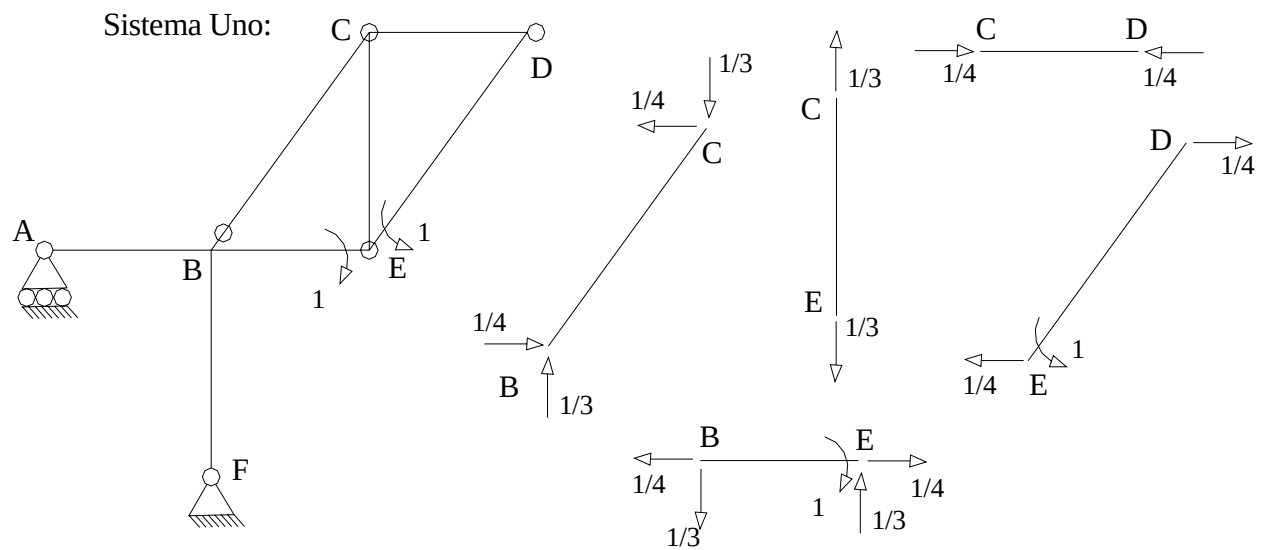


Sistema Cero:

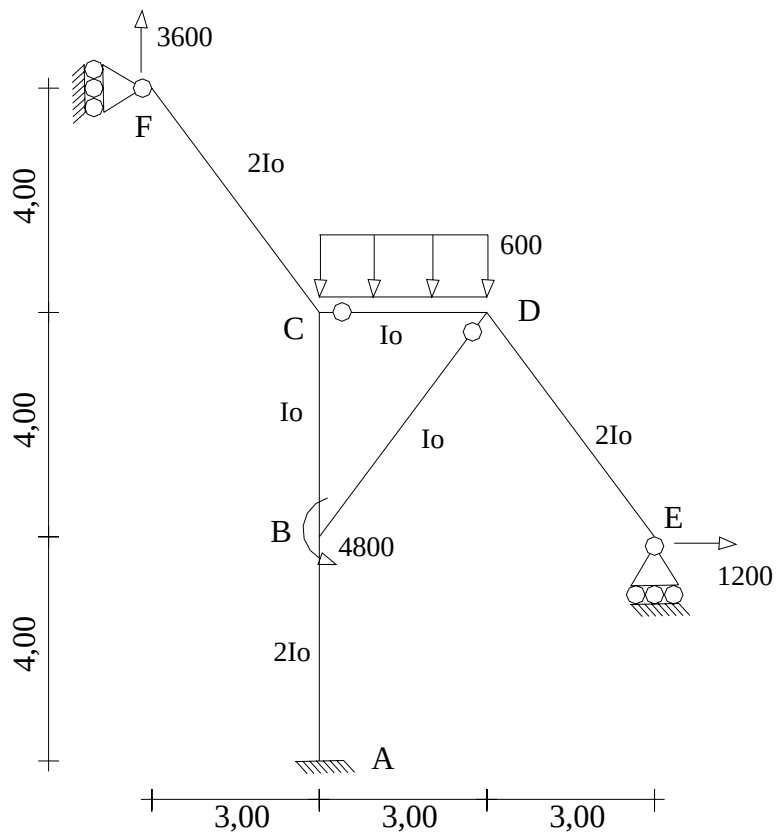


$$+\circlearrowleft \sum M_F = 0 \Rightarrow -2400 + 3 \times 1200 + 3 \times 600 \times 4,5 + 3A_v = 0 \Rightarrow A_v = -3100$$





9.-

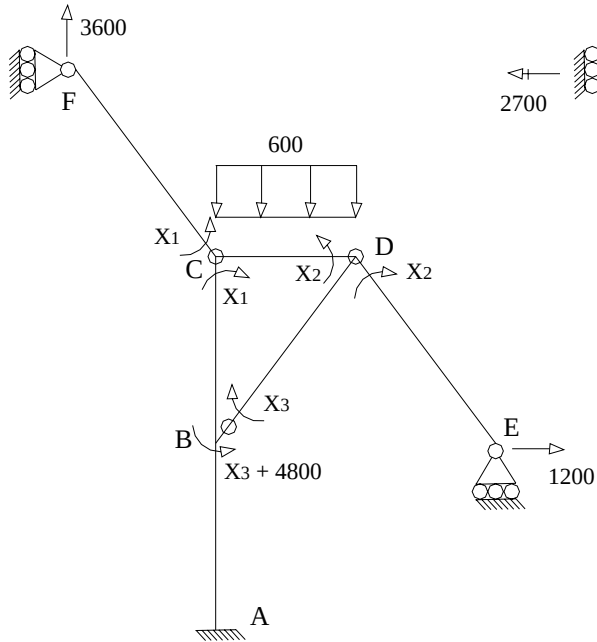


$$\theta_A = 2 \times 10^{-3}$$

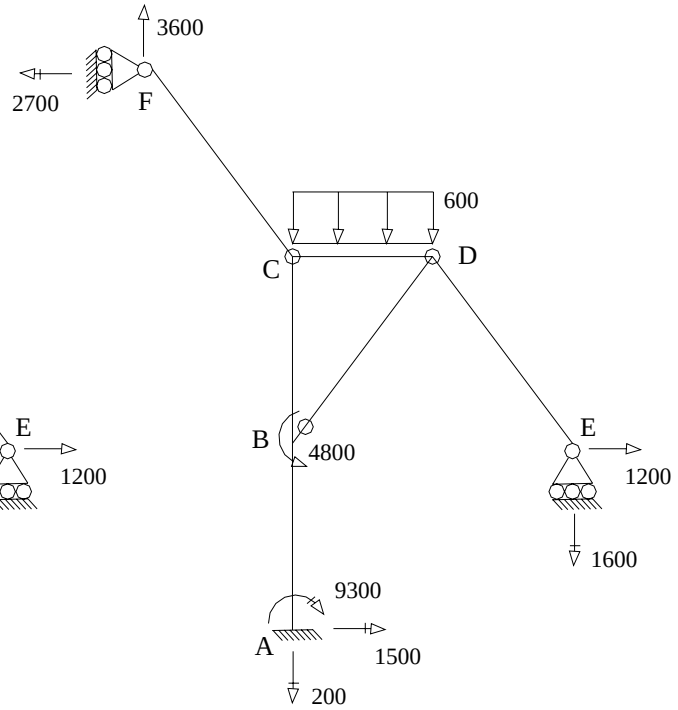
$$\Delta t_{o_{BD}} = 15^\circ \text{C}$$

$$\alpha t = 1 \times 10^{-5}$$

Sistema Primario:



Sistema Cero:

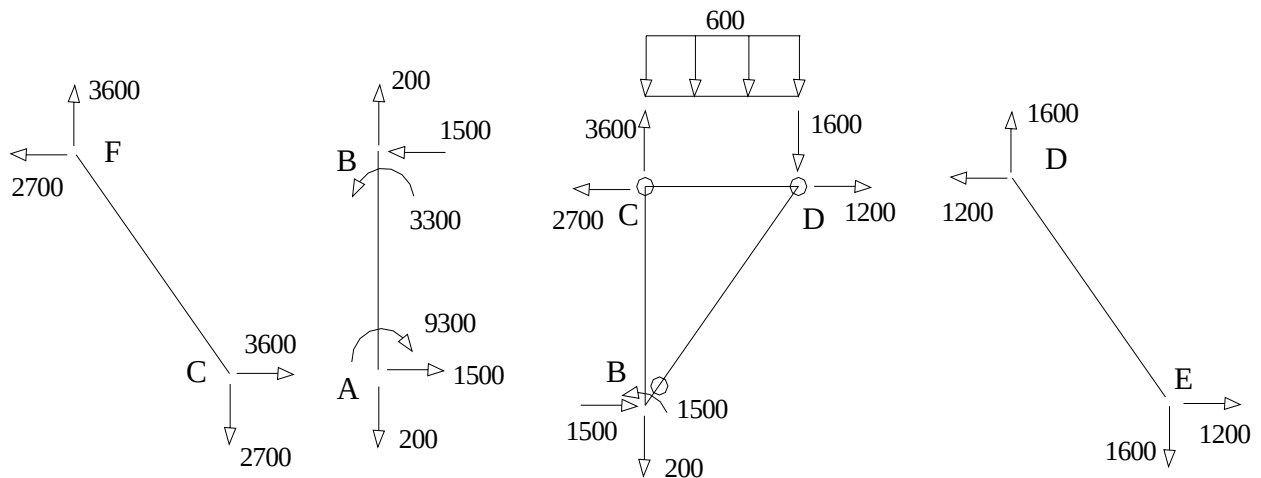


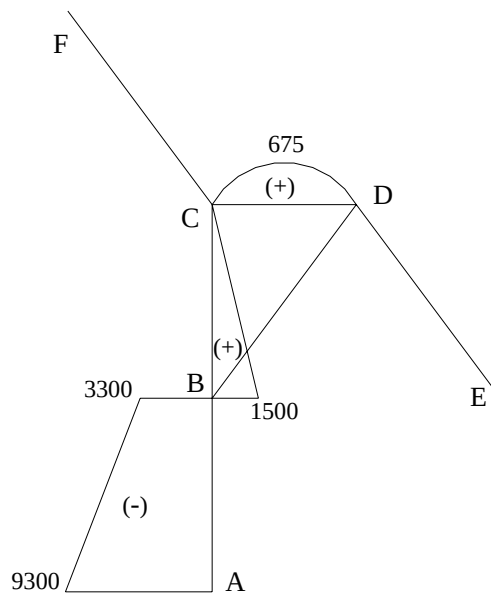
$$+\circlearrowleft \sum M_C^{CF} = 0 \Rightarrow 3 \times 3600 + 4F_h = 0 \Rightarrow F_h = -2700$$

$$+\circlearrowleft \sum M_D^{DE} = 0 \Rightarrow -4 \times 1200 - 3E_v = 0 \Rightarrow E_v = -1600$$

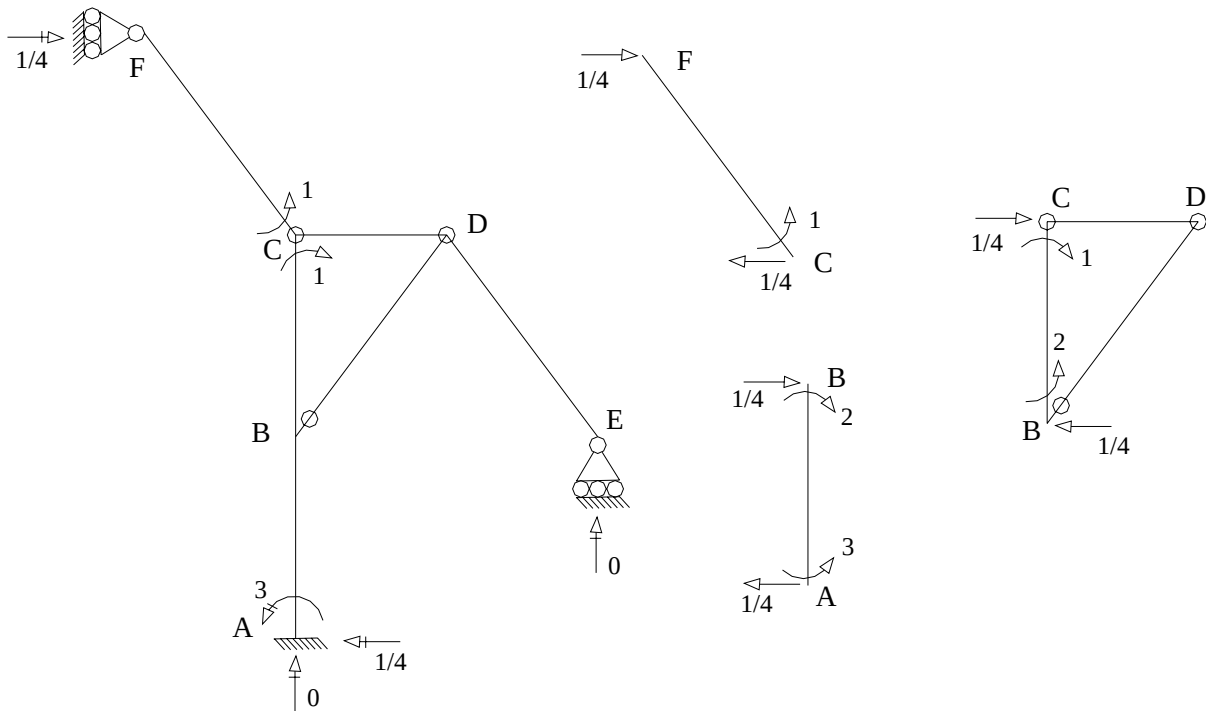
$$+\circlearrowleft \sum M_A = 0 \Rightarrow -4800 + 600 \times 3 \times \frac{3}{2} + 6 \times 1600 + 4 \times 1200 + 3 \times 3600 - 12 \times 2700 + M_A = 0$$

$$M_A = 9300$$



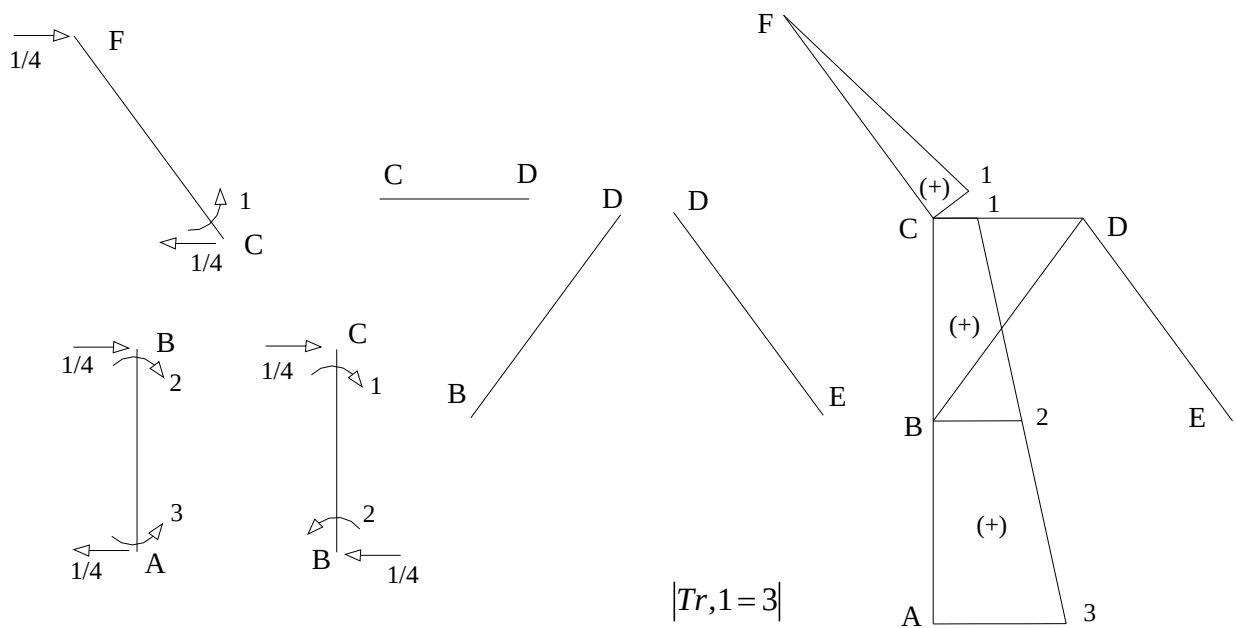


Sistema Uno:

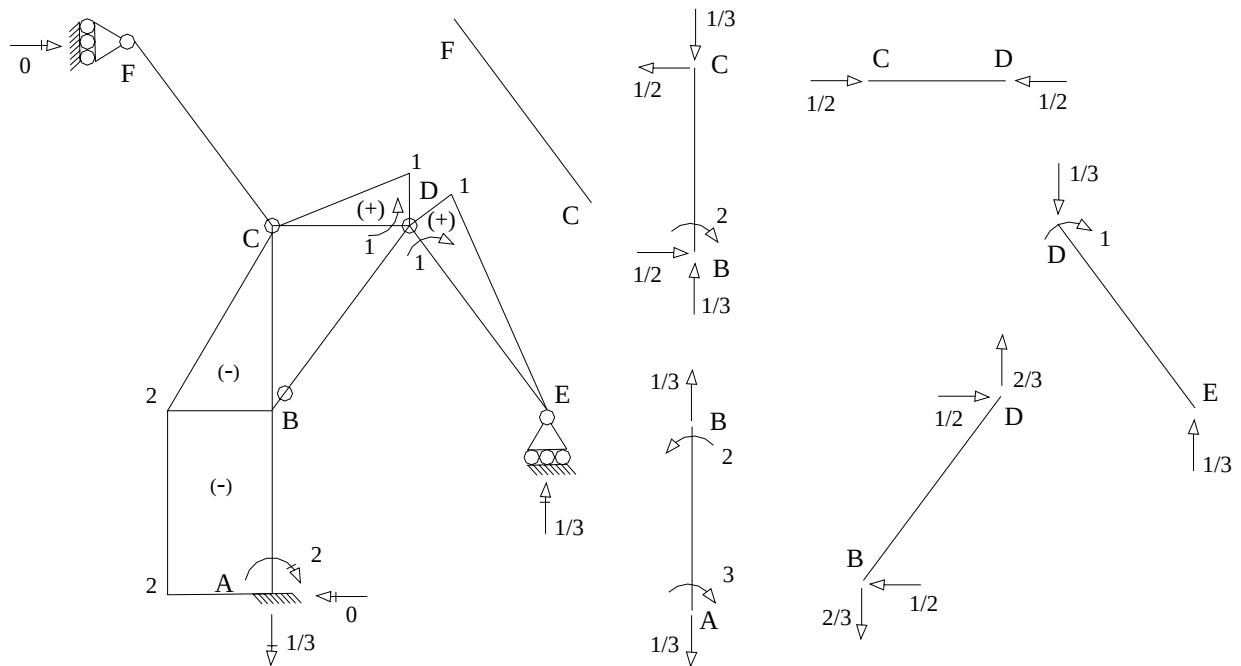


$$+\circlearrowleft \sum M_C^{CF} = 0 \Rightarrow -1 + 4F_h = 0 \Rightarrow F_h = \frac{1}{4}$$

$$+\circlearrowleft \sum M_A = 0 \Rightarrow 12 \times \frac{1}{4} + M_A = 0 \Rightarrow M_A = -3$$



Sistema Dos:

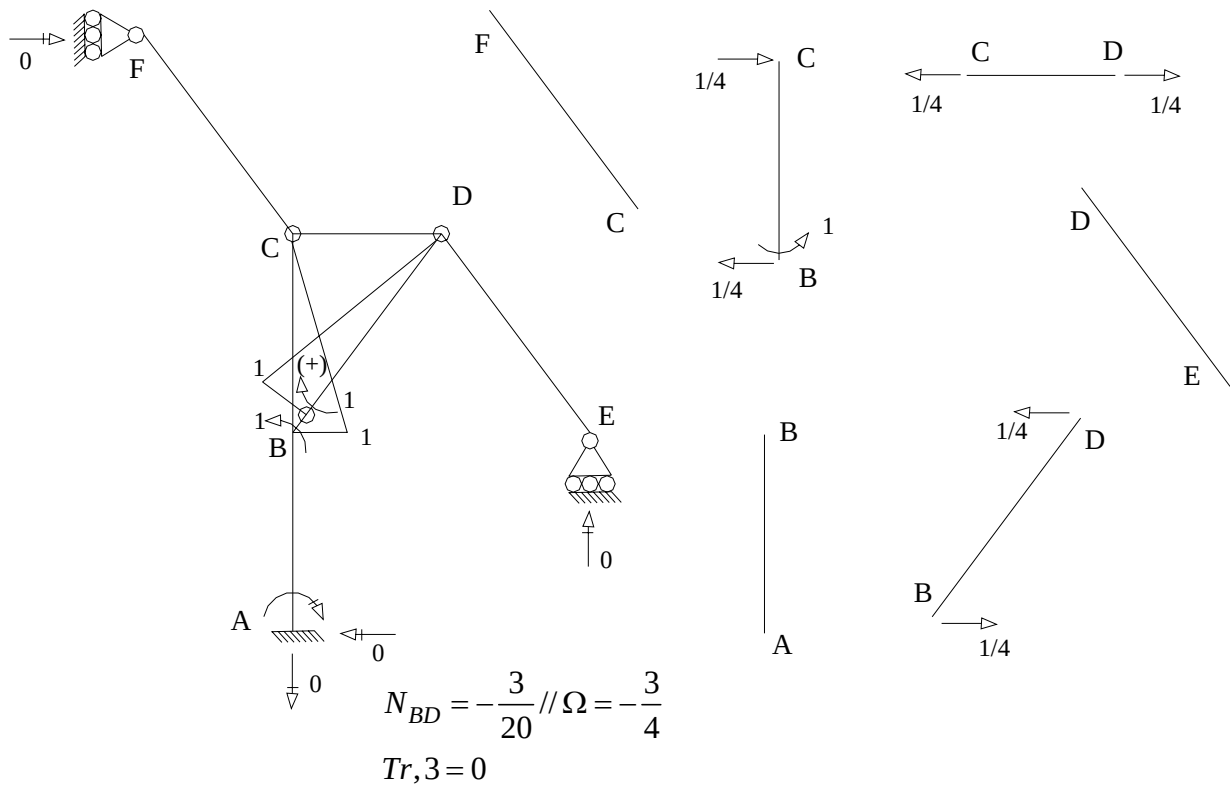


$$\bar{X} = -\frac{1}{2}; \quad \bar{Y} = -\frac{2}{3}; \quad N_{BD} = -\bar{X}\cos\phi - \bar{Y}\sin\phi$$

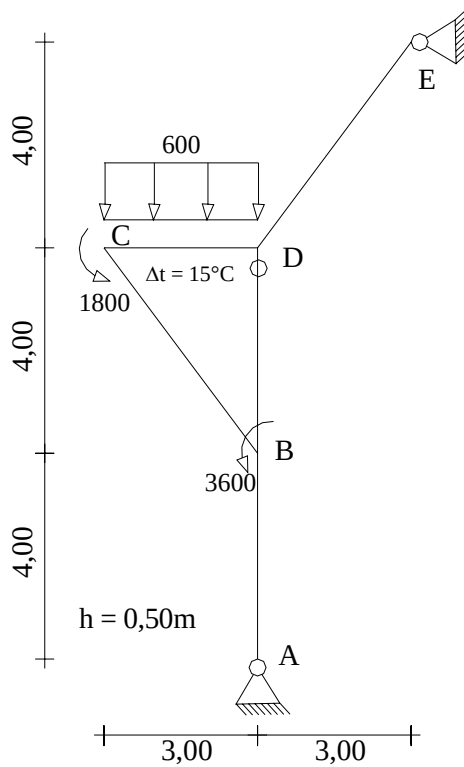
$$N_{BD} = \frac{1}{2}\left(\frac{3}{5}\right) + \frac{2}{3}\left(\frac{4}{5}\right) = \frac{3}{10} + \frac{8}{15} = \frac{45+80}{150}; \quad N_{BD} = \frac{125}{150} = \frac{25}{30} = \frac{5}{6} // \Omega = \frac{25}{6}$$

$$Tr, 1 = 2$$

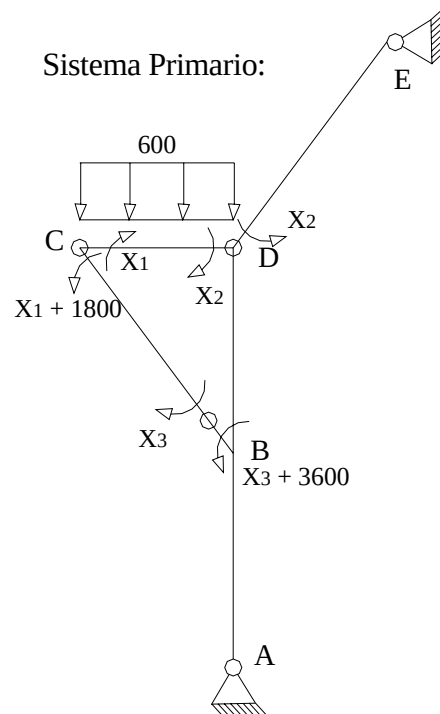
Sistema Tres:



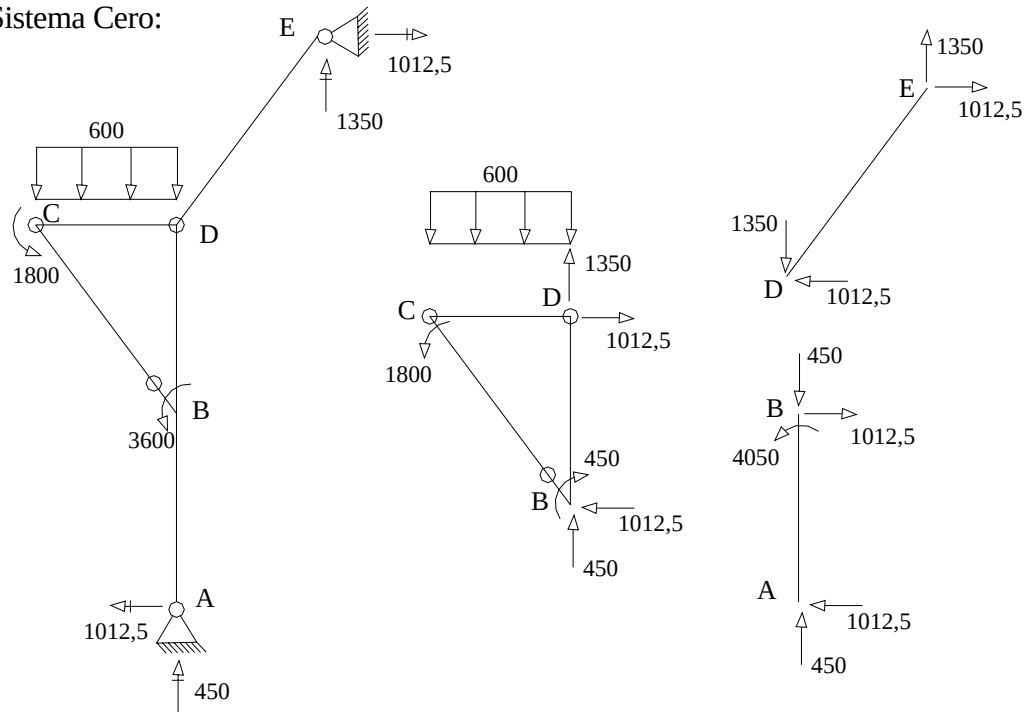
10.-



Sistema Primario:

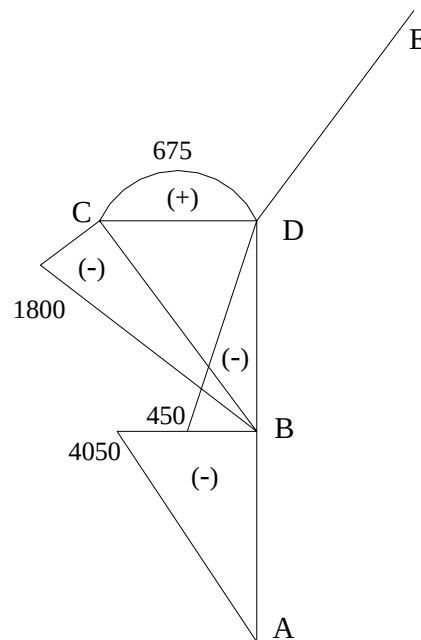


Sistema Cero:

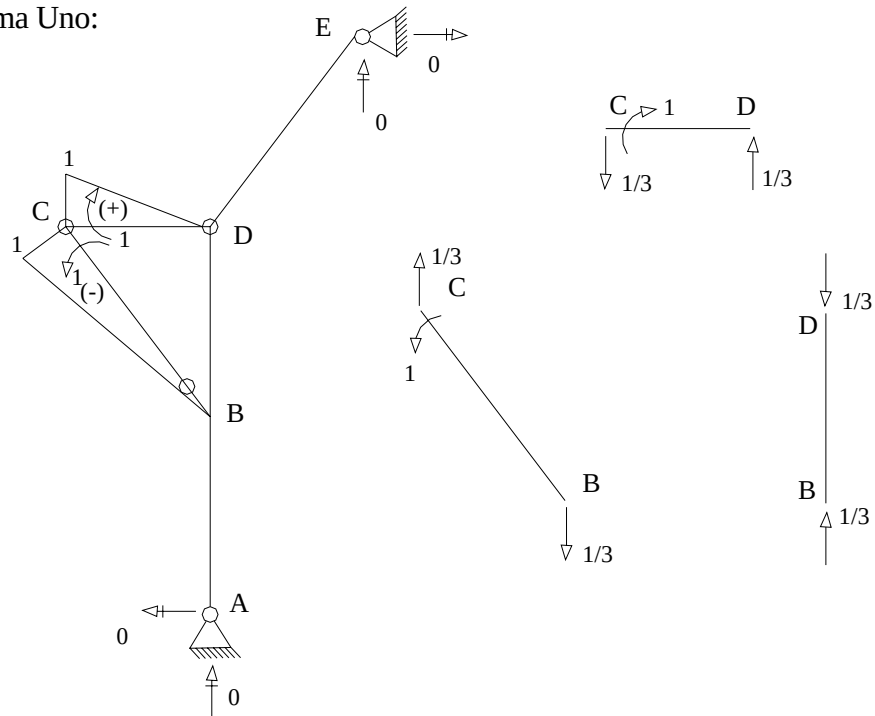


$$+\circlearrowleft \sum M_D^{DCBA} = 0 \Rightarrow -3600 - 1800 - 2700 - 8A_h = 0 \Rightarrow A_h = -1012,5$$

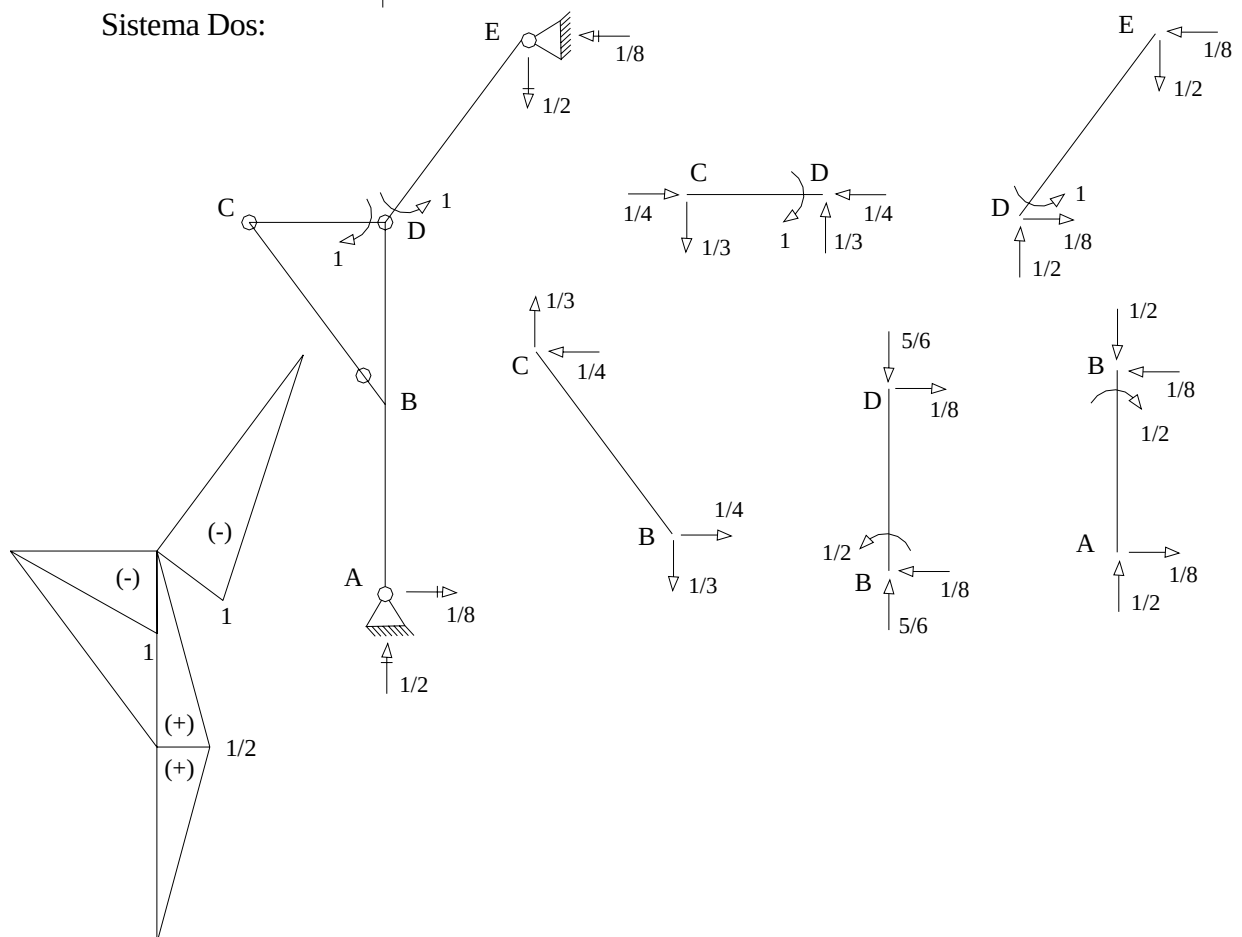
$$+\circlearrowleft \sum M_D^{DE} = 0 \Rightarrow 4 \times 1012,5 - 3E_v = 0 \Rightarrow E_v = 1350$$



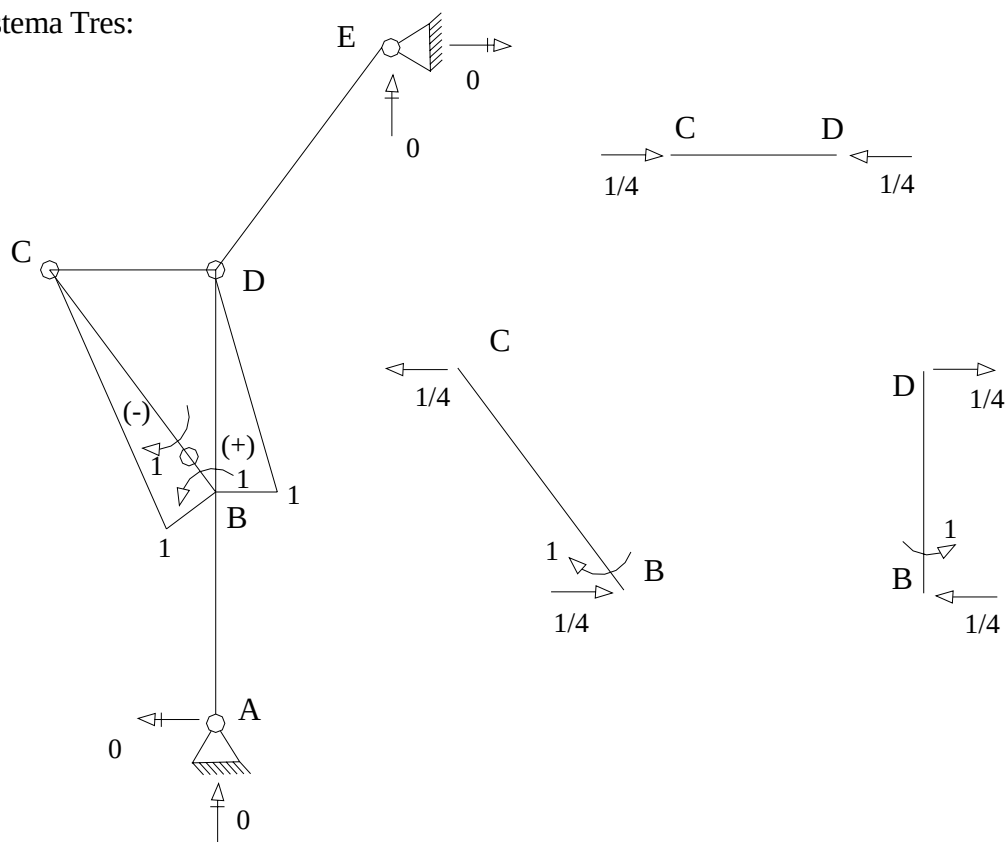
Sistema Uno:



Sistema Dos:



Sistema Tres:



$$D_{10} = \frac{1}{3} \times \frac{675 \times 3}{EI_o} + \frac{1}{3} \times \frac{1800 \times 5}{EI_o} = \frac{3675}{EI_o}$$

$$D_{20} = -\frac{1}{3} \times \frac{675 \times 3}{EI_o} - \frac{1}{6} \times \frac{1}{2} \times \frac{450}{EI_o} \times 4 - \frac{1}{6} \times \frac{1}{2} \times \frac{4050}{EI_o} \times 4 = -\frac{2175}{EI_o}$$

$$D_{30} = \frac{1}{6} \times \frac{1800 \times 5}{EI_o} - \frac{1}{3} \times \frac{450 \times 4}{EI_o} = \frac{900}{EI_o}$$

$$d_{11} = (4+5) \frac{1}{3EI_o} = \frac{3}{EI_o};$$

$$d_{22} = \frac{1}{3EI_o} (5+3+1+1) = \frac{10}{3EI_o}$$

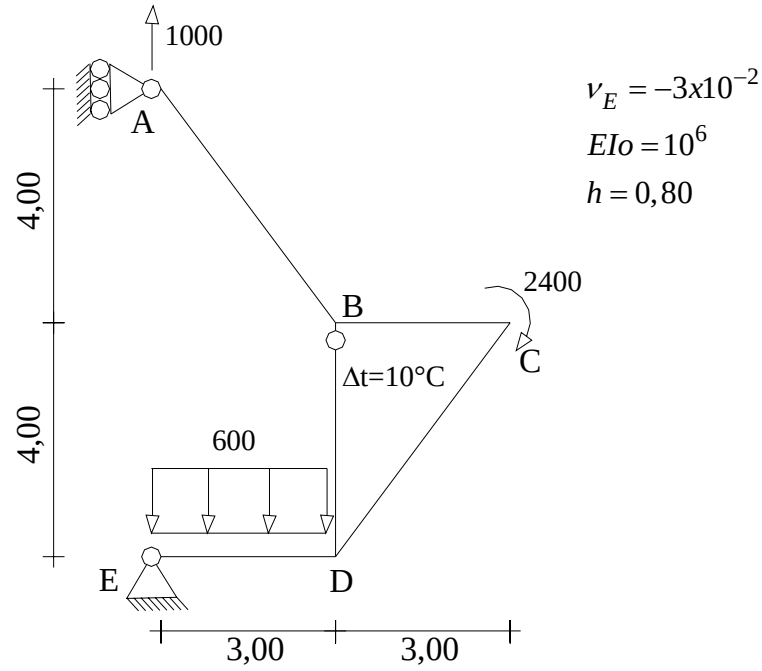
$$d_{33} = (4+5) \frac{1}{3EI_o} = \frac{3}{EI_o};$$

$$d_{12} = d_{21} = -\frac{3}{6EI_o} = -\frac{1}{3EI_o}$$

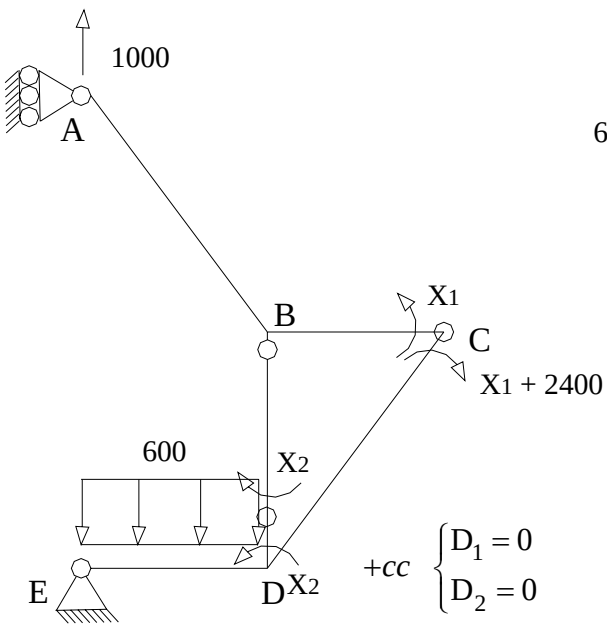
$$d_{13} = d_{31} = \frac{5}{6EI_o};$$

$$d_{23} = d_{32} = \frac{1}{3} \times \frac{1}{2} \times \frac{1 \times 4}{EI_o} = -\frac{2}{3EI_o}$$

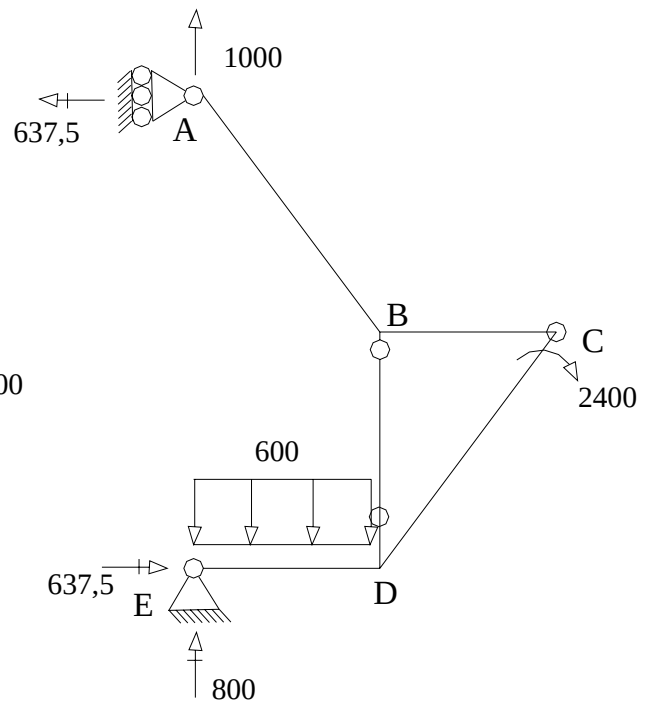
11.-



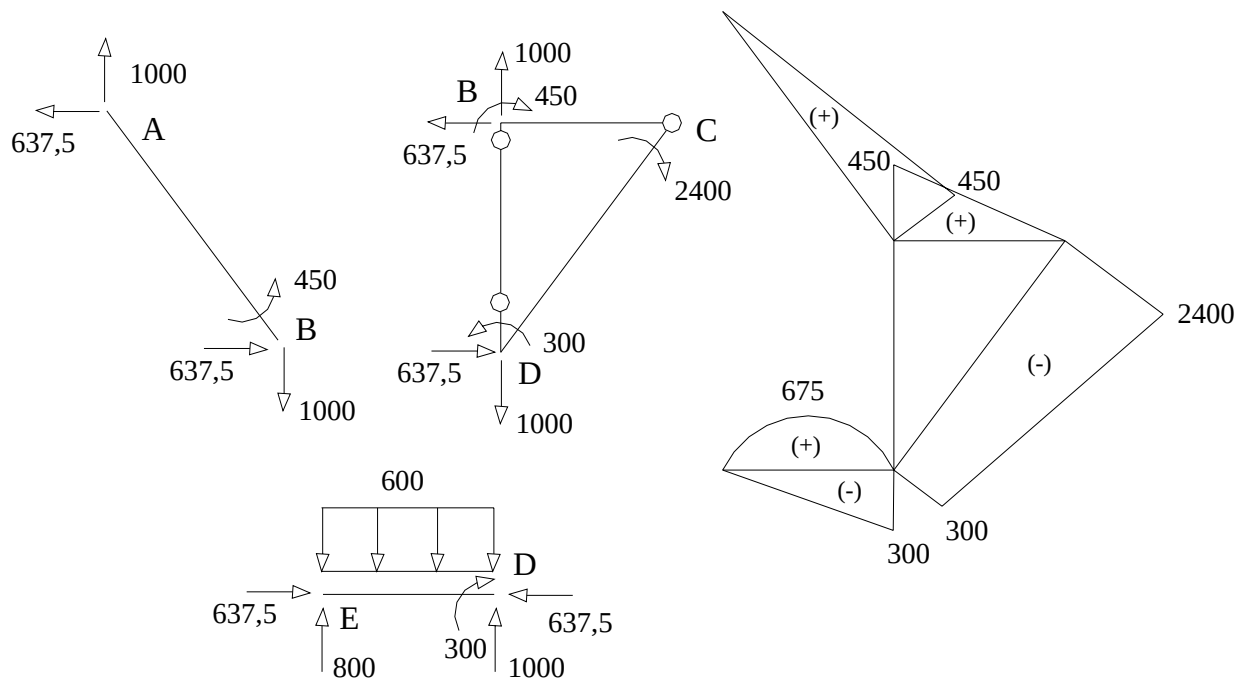
Sistema Primario:



Sistema Cero:



$$+\circlearrowleft \sum M_E = 0 \Rightarrow 2400 + 600 \times \frac{3^2}{2} + 8A_h = 0 \Rightarrow A_h = -637,5$$



$$D_{10} = \frac{1}{6} \times \frac{450 \times 3}{EI_o} + \frac{5}{6} (2400 \times 1 \times 2 + 300 \times 1) + \left(-\frac{4}{3} + \frac{4}{3}\right) \times 10^{-5} \times 5 \times \frac{10^6}{EI_o} +$$

$$+ \left[\frac{3}{2} \times 10 - \frac{5}{2} (-10)\right] \times 10^{-5} \times \frac{10^6}{EI_o} \times \frac{1}{0,8} = \frac{4975}{EI_o}$$

$$D_{20} = \frac{5}{6} (300 \times 1 \times 2 + 2400 \times 1) + \left(-\frac{3}{4} + \frac{3}{4}\right) \times 10^{-5} \times 5 \times \frac{10^6}{EI_o} +$$

$$+ \left[2 \times 10 - \frac{5}{2} (-10)\right] \times 10^{-5} \times \frac{10^6}{EI_o} \times \frac{1}{0,8} = \frac{3062,5}{EI_o}$$

$$d_{11} = (3+5) \frac{1}{3EI_o} = \frac{8}{3EI_o}; \quad d_{22} = \frac{1}{3EI_o} (5+4) = \frac{3}{EI_o}$$

$$d_{12} = d_{21} = \frac{5}{6EI_o}$$